1. Let $x = 2^{\log 3}$ and $y = 3^{\log 2}$ where base of the logarithm is 10, then which one of the following holds good?

यदि $x = 2^{\log 3}$ व $y = 3^{\log 2}$ जहाँ लघुगणक का आधार 10 हो, तो निम्न में से कौन सत्य होगा?

(A) $2x < y$ (B) $2y < x$ (C) $3x = 2y$ (D) $y = x$

Ans. D

Sol. $x = 2^{\log 3}$ and $y = 3^{\log 2}$

taking log with base 10

$\log_{10} x = \log 3 \cdot \log 2$ ...(1)

taking log with base 10

$\log_{10} y = \log 2 \cdot \log 3$ ...(2)

from (1) & (2)

$\log_{10} x = \log_{10} y$

$\Rightarrow x = y$

2. If $\log_y x + \log_x y = 7$, then the value of $(\log_y x)^2 + (\log_x y)^2$, is :

यदि $\log_y x + \log_x y = 7$ तो $(\log_y x)^2 + (\log_x y)^2$ का मान होगा :

(A) 43 (B) 45 (C) 47 (D) 49

Ans. C

Sol. if $\log_y x + \log_x y = 7$

Squaring both sides

$(\log_y x + \log_x y)^2 = 49$

$(\log_y x)^2 + (\log_x y)^2 + 2 \cdot \log_y x \cdot \log_x y = 49$

$(\log_y x)^2 + (\log_x y)^2 + 2 = 49$

$(\log_y x)^2 + (\log_x y)^2 = 49 - 2 = 47$

3. The value of 'a' for which $\frac{\log_a 7}{\log_6 7} = \log_\pi 36$ holds good, is :

यदि $\frac{\log_a 7}{\log_6 7} = \log_\pi 36$ तो a का मान होगा :

(A) $\frac{1}{\pi}$ (B) π^2 (C) $\sqrt{\pi}$ (D) 2

Ans. C

Sol. $\log_a 7 \cdot \log_7 6 = \log_\pi 36$

$\log_a 6 = 2\log_\pi 6 = \dfrac{2\log_{\sqrt{\pi}} 6}{\log_{\sqrt{\pi}} \pi} = \log_{\sqrt{\pi}} 6$. Hence a = $\sqrt{\pi}$ Ans.

4. The expression $\sqrt{12+6\sqrt{3}} + \sqrt{12-6\sqrt{3}}$ simplifies to :

व्यंजक $\sqrt{12+6\sqrt{3}} + \sqrt{12-6\sqrt{3}}$ सरल करने पर विघटित होगा :

(A) 4 (B) $2\sqrt{3}$ (C) $3\sqrt{3}$ (D) 6

Ans. D

Sol. $\because 12+6\sqrt{3} = 12+2\times 3\sqrt{3}$

$= 3^2 + \sqrt{3}^2 + 2\cdot 3\cdot\sqrt{3}$

$= \left(3+\sqrt{3}\right)^2$

And $12-6\sqrt{3} = \left(3-\sqrt{3}\right)^2$

$\therefore \sqrt{12+6\sqrt{3}} + \sqrt{12-6\sqrt{3}}$

$= 3+\sqrt{3}+3-\sqrt{3}$

$= 6$

5. If $\log_2\left(\log_3\left(\log_4(x)\right)\right) = 0$ and $\log_3\left(\log_4\left(\log_2(y)\right)\right) = 0$ and $\log_4\left(\log_2\left(\log_3(z)\right)\right) = 0$ then the sum of x, y and z is :

यदि $\log_2\left(\log_3\left(\log_4(x)\right)\right) = 0$ और $\log_3\left(\log_4\left(\log_2(y)\right)\right) = 0$ और $\log_4\left(\log_2\left(\log_3(z)\right)\right) = 0$ तो x, y व z का योग होगा :

(A) 89 (B) 58 (C) 105 (D) 50

Ans. A

Sol. if $\log_2(\log_3(\log_4(x))) = 0$

$\log_3(\log_4(x)) = 2^\circ = 1$

$\log_4 x = 3^1 = 3$

$x = 4^3 = 64$

and $\log_3(\log_4(\log_2(y))) = 0$

$\log_4(\log_2(y)) = 3^\circ = 1$

$\log_2 y = 4^1$

$\Rightarrow y = 2^4 = 16$

And $\log_4(\log_2(\log_3(z))) = 0$

$\log_2\log_3 z = 4^\circ = 1$

$\log_3 z = 2^1$
$z = 3^2$
$z = 9$
Now $x + y + z = 64 + 16 + 9 = 89$

6. $\log_{10}(\log_2 3) + \log_{10}(\log_3 4) + \log_{10}(\log_4 5) + ... + \log_{10}(\log_{1023} 1024)$ simplifies to :

(A) a composite (B) a prime number

(C) rational which is not an integer (D) an integer

व्यजक $\log_{10}(\log_2 3) + \log_{10}(\log_3 4) + \log_{10}(\log_4 5) + ... + \log_{10}(\log_{1023} 1024)$ सरल होने पर बनेगा :

(A) एक मिश्रित संख्या (B) एक अभाज्य संख्या

(C) परिमेय जो कि पूर्णांक नही है (D) एक पूर्णांक

Ans. D

Sol. $\log_{10}(\log_2 3) + \log_{10}(\log_3 4) + \log_{10}(\log_4 5) + ... + \log_{10}(\log_{1023} 1024)$
$= \log_{10}[\log_2 3 \times \log_3 4 \times \log_4 5 \times ..... \times \log_{1023} 1024]$

$$= \log_{10}\left[\frac{\log 3}{\log 2} \times \frac{\log 4}{\log 3} \times \frac{\log 5}{\log 4} \times ..... \times \frac{\log 1024}{\log 1023}\right]$$

$= \log_{10}[\log_2 1024]$
$= \log_{10} 10 = 1$
An integer

7. Number of digits in $4^{16} \cdot 5^{25}$ is : (use $\log_{10} 2 = 0.3010$)

$4^{16} \cdot 5^{25}$ में कुल कितने अंक होगे ($\log_{10} 2 = 0.3010$)

(A) 27 (B) 28 (C) 29 (D) 30

Ans. B

Sol. $N = 2^{32} \cdot 5^{25} = 2^7(2 \cdot 5)^{25} = 2^7 \cdot 10^{25}$
hence $\log_{10} N = 25 + 7 \log_{10} 2 = 25 + 7 \times 0.3010 = 25 + 2.107 = 27.107$
$\Rightarrow$ N is a 28 digit number

8. If $\log_a b + \log_b c + \log_c a$ vanishes where a, b and c are positive reals different than unity then the value of $(\log_a b)^3 + (\log_b c)^3 + (\log_c a)^3$ is :

(A) an odd prime (B) an even prime (C) an odd composite (D) an irrational number

यदि $\log_a b + \log_b c + \log_c a$ विलोपित हो जाता है, जहां पर a, b व c धनात्मक वास्तविक व इकाई से भिन्न संख्याऐं हो तो $(\log_a b)^3 + (\log_b c)^3 + (\log_c a)^3$ का मान होगा :

(A) एक विषम अभाज्य (B) एक सम अभाज्य (C) एक विषम मिश्रित संख्या (D) एक अपरिमेय संख्या

Ans. A

Sol. if $\log_a b + \log_b c + \log_c a = 0$
then $(\log_a b)^3 + (\log_b c)^3 + (\log_c a)^3$
$= 3 \log_a b \cdot \log_b c \cdot \log_c a$

$$= 3\frac{\log b}{\log a} \times \frac{\log c}{\log b} \times \frac{\log a}{\log c} = 3$$

9. If $\log_{10}(ab^3) = \log_{10}(a^2b) = 1$, then the value of $\log_{10}(ab)$ is equal to :

यदि $\log_{10}(ab^3) = \log_{10}(a^2b) = 1$, तो $\log_{10}(ab)$ का मान होगा :

(A) $\frac{5}{3}$ (B) $\frac{3}{5}$ (C) $\frac{2}{5}$ (D) $\frac{1}{5}$

Ans. B

Sol. We have $\log_{10}(ab^3) = 1 \quad \Rightarrow \quad ab^3 = 10$ _____(1)
Also $\log_{10}(a^2b) = 1 \quad \Rightarrow \quad a^2b = 10$ _____(2)
$\therefore$ From equation (1) and equation (2), we get
$ab^3 = a^2b$
$\Rightarrow \quad ab(b^2 - a) = 0$
$\therefore \; b^2 = a$ (As $a \neq 0, b \neq 0$)
Now on putting $a = b^2$ in equation (1), we get

$$b^5 = 10$$

$$\therefore \quad b = 10^{\frac{1}{5}} \quad \text{and} \quad a = 10^{\frac{2}{5}}$$

Hence $\log_{10}(ab) = \log_{10}\left(10^{\frac{2}{5}} \cdot 10^{\frac{1}{5}}\right) = \log_{10}\left(10^{\frac{3}{5}}\right) = \frac{3}{5}$ Ans.

10. If $x = \left(2^{\sqrt{5}}\right)\left(5^{\sqrt{2}}\right)$, then $\log_{10} x = \left(\sqrt{A} - \sqrt{B}\right)(\log_{10}2) + \sqrt{B}$. The value of $(A + B)$ equals :

यदि $x = \left(2^{\sqrt{5}}\right)\left(5^{\sqrt{2}}\right)$, तो $\log_{10} x = \left(\sqrt{A} - \sqrt{B}\right)(\log_{10}2) + \sqrt{B}$ उपरोक्त रूप में व्यक्त जहाँ पर $(A + B)$ बराबर होगा :

(A) 7 (B) 9 (C) 11 (D) 13

Ans. A

Sol. $x = 2^{\sqrt{5}} \cdot 5^{\sqrt{2}}$

$$\log_{10} x = \sqrt{5}\,\log_{10}2 + \sqrt{2}\,[1 - \log_{10}2]$$

or $\log_{10}x = \left(\sqrt{5}-\sqrt{2}\right)\log_{10}2 + \sqrt{2} = \left(\sqrt{A}-\sqrt{B}\right)\log_{10}2 + \sqrt{2}$

$A = 5;\ B = 2 \Rightarrow A + B = 7$ Ans.

11. If $\sum_{r=0}^{n-1}\log_2\left(\frac{r+2}{r+1}\right) = \prod_{r=10}^{99}\log_r(r+1)$, then 'n' is equal to :

यदि $\sum_{r=0}^{n-1}\log_2\left(\frac{r+2}{r+1}\right) = \prod_{r=10}^{99}\log_r(r+1)$ है, तो 'n' का मान होगा :

(A) 4 (B) 3 (C) 5 (D) 6

Ans. B

Sol. $\text{L.H.S.} = \sum_{r=0}^{n-1}\log_2\left(\frac{r+2}{r+1}\right)$

$= \log_2\left(\frac{2}{1}\right)+\log_2\left(\frac{3}{2}\right)+\log\left(\frac{4}{3}\right)+.....+\log_2\left(\frac{n}{n-1}\right)+\log_2\left(\frac{n+1}{n}\right)$

$= \log_2\left(\frac{2}{1}\times\frac{3}{2}\times\frac{4}{3}\times.....\times\frac{n}{n-1}\times\frac{n+1}{n}\right)$

$= \log_2(n+1)$ .... (1)

Now R.H.S. $= \prod_{r=10}^{99}\log_r(r+1) = \log_{10}11 \times \log_{11}12 \times \log_{12}13 \times ..... \times \log_{98}99 \times \log_{99}100$

$= \frac{\log 11}{\log 10}\times\frac{\log 12}{\log 11}\times\frac{\log 13}{\log 12}\times.....\times\frac{\log 99}{\log 98}\times\frac{\log 100}{\log 99}$

$= \log_{10}100 = 2$ ... (2)

$\therefore$ From (1) & (2), we get

$\log_2(n+1) = 2$

$\Rightarrow$ $n + 1 = 4$

$\therefore$ $n = 3$ Ans.

12. Let $3^a = 4$, $4^b = 5$, $5^c = 6$, $6^d = 7$, $7^e = 8$ and $8^f = 9$. The value of the product (abcdef), is :

माना $3^a = 4$, $4^b = 5$, $5^c = 6$, $6^d = 7$, $7^e = 8$ तथा $8^f = 9$ है, तो गुणनफल (abcdef) का मान होगा :

(A) 1 (B) 2 (C) $\sqrt{6}$ (D) 3

Ans. B

Sol. $3^a = 4;\ a = \log_3 4;$ |||ly $b = \log_4 5$ etc.

hence $abcdef = \log_3 4 \cdot \log_4 5 \cdot \log_5 6 \cdot \log_6 7 \cdot \log_7 8 \cdot \log_8 9 = \log_3 9 = 2 \Rightarrow$ 2 Ans.

13. If $\log_9 x + \log_4 y = \frac{7}{2}$ and $\log_9 x - \log_8 y = -\frac{3}{2}$, then $x + y$ equals :

यदि $\log_9 x + \log_4 y = \frac{7}{2}$ और $\log_9 x - \log_8 y = -\frac{3}{2}$ तो $x + y$ का मान होगा :

(A) 35 (B) 41 (C) 67 (D) 73

Ans. C

Sol. If $\log_9 x + \log_4 y = \frac{7}{2}$ and $\log_9 x - \log_8 y = \frac{-3}{2}$

$\frac{1}{2}\log_3 x + \frac{1}{2}\log_2 y = \frac{7}{2}$ $\frac{1}{2}\log_3 x - \frac{1}{3}\log_2 y = \frac{-3}{2}$

$\log_3 x + \log_2 y = 7$ ...(1) $\frac{1}{2}(7 - \log_2 y) - \frac{1}{3}\log_2 y = \frac{-3}{2}$

$\Rightarrow \log_3 x = 7 - \log_2 y$ $\frac{7}{2} - \frac{1}{2}\log_2 y - \frac{1}{3}\log_2 y = \frac{-3}{2}$

Now $\log_3 x = 7 - 6 = 1$ $-\log_2 y \times \left\{\frac{1}{2} + \frac{1}{3}\right\} = -\left(\frac{3}{2} + \frac{7}{2}\right)$

$x = 3^1$ $\frac{5}{6} \times \log_2 y = 5$

$x = 3$ $\log_2 y = 6$

$y = 2^6 = 64$

then $x + y = 3 + 64 = 67$

14. If $x = \log_k b = \log_b c = \frac{1}{2}\log_c d$ then $\log_k d$ equals :

यदि $x = \log_k b = \log_b c = \frac{1}{2}\log_c d$ तो $\log_k d$ का मान होगा :

(A) $2x^3$ (B) $\frac{x^3}{2}$ (C) $2x^8$ (D) $6x$

Ans. A

Sol. if $x = \log_k b = \log_b c = \frac{1}{2}\log_c d$

$\log_k b = x$ ...(1)

$\log_b c = x$ ...(2)

$\frac{1}{2}\log_c d = x$

$\log_c d = 2x$ ...(3)

by (1) × (2) × (3)

$\log_k b \times \log_b c \times \log_c d = x \times x \times 2x$

$\log_k d = 2x^3$

ONE OR MORE THAN CORRECT TYPE

15. The expression, $\log_p \log_p \underbrace{\sqrt[p]{\sqrt[p]{\sqrt[p]{.....\sqrt[p]{p}}}}}_{\text{n radical sign}}$ where $p \geq 2$, $p \in N$, when simplified is :

(A) independent of p, but dependent on n
(B) independent of n, but dependent on p
(C) dependent on both p and n
(D) negative

यदि $p \geq 2$, $p \in N$ तो व्यंजक $\log_p \log_p \underbrace{\sqrt[p]{\sqrt[p]{\sqrt[p]{.....\sqrt[p]{p}}}}}_{\text{n radical sign}}$ सरल करने पर होगा :

(A) p से स्वतंत्र परन्तु n से नहीं
(B) n से स्वतंत्र परन्तु p से नहीं
(C) n तथा p दोनों से स्वतंत्र
(D) ऋणात्मक होगा

Ans. AD

Sol. The expression

$\log_p \log_p \underbrace{\sqrt[p]{\sqrt[p]{\sqrt[p]{.....\sqrt[p]{p}}}}}_{\text{n radical sign}}$ where $p \geq 2$, $p \in N$

$= \log_p \log_p p^{\frac{1}{p^n}}$

$= \log_p \frac{1}{p^n} \times 1$

$= -n \times 1$

$= -n$

16. If $\log_4 5 = x$ and $\log_5 6 = y$ then :

यदि $\log_4 5 = x$ तथा $\log_5 6 = y$ तो :

(A) $\log_4 6 = xy$ (B) $\log_6 4 = xy$ (C) $\log_3 2 = \frac{1}{2xy-1}$ (D) $\log_2 3 = \frac{1}{2xy-1}$

Ans. AC

Sol. $xy = \log_4 5 \cdot \log_5 6 = \log_4 6 \quad \Rightarrow \quad$ (A) is correct

Also $xy = \frac{\log_2 6}{\log_2 4} \quad \Rightarrow \quad 2xy = 1 + \log_2 3$

$2xy - 1 = \log_2 3$

$\Rightarrow \log_3 2 = \frac{1}{2xy-1} \quad \Rightarrow \quad$ (C) is correct

17. Select the correct statement :

(A) $\log_3 19 \cdot \log_{1/7} 3 \cdot \log_4\left(\frac{1}{7}\right) < 2$

(B) The equation $\log_{1/3}(x^2 + 8) = -2$ has two real solutions

(C) Let $N = \log_2 15 \cdot \log_{1/6} 2 \cdot \log_3\left(\frac{1}{6}\right)$. The greatest integer which is less than or equal to N is 3

(D) The equation $\log_4 x + \log_4(x+2) = \log_4(3x)$ has no prime solution

सही कथन चुने :

(A) $\log_3 19 \cdot \log_{1/7} 3 \cdot \log_4\left(\frac{1}{7}\right) < 2$

(B) समीकरण $\log_{1/3}(x^2 + 8) = -2$ के दो वास्तविक हल होगें

(C) यदि $N = \log_2 15 \cdot \log_{1/6} 2 \cdot \log_3\left(\frac{1}{6}\right)$ तो महत्तम पूर्णांक जो N के बराबर या N से छोटा 3 है

(D) समीकरण $\log_4 x + \log_4(x+2) = \log_4(3x)$ का कोई अभाज्य हल नहीं करें

Ans. BD

Sol. (A) $\log_3 19 \cdot \log_{\frac{1}{7}} 3 \cdot \log_4\left(\frac{1}{7}\right)$

$= \log_{\frac{1}{7}} 19\{-\log_4 7\}$

$= -\{\log_7 19\} \times \{-\log_4 7\}$

$= \log_7 19 \times \log_4 7$

$= \log_4 19$

$= \frac{\log_{10} 19}{\log_{10} 2} = \frac{1.2787}{2 \times 0.3010} = \frac{12787}{6020} = 2.1240 \not< 2$ Incorrect

(B) The equation

$\log_{\frac{1}{3}} \left(x^2 + 8\right) = -2$

$x^2 + 8 = \left(\frac{1}{3}\right)^{-2}$

$x^2 + 8 = 9$

$x^2 = 1$

$x = \pm 1$

two real solutions. Correct

(C) Let $N = \log_2 15 \cdot \log_{\frac{1}{6}} 2 \cdot \log_3 \left(\frac{1}{6}\right)$

$N = \log_2 15 \cdot \log_3 2$

$N = \log_3 15$

$N = \log_3 3 + \log_3 5$

$N = 1 + \frac{\log_{10} 5}{\log_{10} 3}$

$N = 1 + \frac{0.6987}{0.4771}$

$N = 1 + 1.4644$

$N = 2.4644$ (Not an integer) $\not\leq 3$ Incorrect

(D) $\log_4 x + \log_4 (x + 2) = \log_4 (3x)$

$\Rightarrow x \cdot (x + 2) = 3x$

$x^2 + 2x - 3x = 0$

$x^2 - x = 0$

$x = 0, 1$ Correct

18. If logarithm base is 10, then the equation, $\sqrt{\log(-x)} = \log \sqrt{x^2}$.

यदि लघुगणक आधार 10 है तो समीकरण $\sqrt{\log(-x)} = \log \sqrt{x^2}$ को हल करें।

Ans. $x = -1$ and $x = -10$

Sol. $\sqrt{\log(-x)} = \log \sqrt{x^2} \Rightarrow \sqrt{\log(-x)} = \log|x|$

Case-I : if $x < 0$

$\sqrt{\log(-x)} = \log(-x)$

Case-II : $x > 0$ (equality couldn't hold)

In this case LHS does not exist for $x > 0$

$\log(-x) - \log^2(-x) = 0$
$\log(-x)\{1 - \log(-x)\} = 0$
if $\log(-x) = 0$
$-x = 1$
$x = -1$
if $1 - \log(-x) = 0$
$\log(-x) = 1$
$-x = 10$
$x = -10$
$\therefore$ value of $x = -1, -10$

19. Compute the value of $81^{\frac{1}{\log_5 3}} + 27^{\log_9 36} + 3^{\frac{4}{\log_7 9}}$

$81^{\frac{1}{\log_5 3}} + 27^{\log_9 36} + 3^{\frac{4}{\log_7 9}}$ का मान ज्ञात कीजिए।

Ans. 890

Sol. $81^{\frac{1}{\log_5 3}} = 3^{4\log_3 5} = 5^4 = 625$ _____(1)

$27^{\log_9 36} = 3^{3\log_{3^2} 36} = 3^{\log_3 (36)^{3/2}} = 6^{2\times\frac{3}{2}} = 216$ _____(2)

$3^{\frac{4}{\log_7 9}} = 3^{4\log_9 7} = 3^{2\log_3 7} = 49$ _____(3)

$625 + 216 + 49 = 890$

MATCH THE COLUMN

20. Match the following Columns :

Following expressions simplifies to

Column-I	**Column-II**
A. $\frac{\log_2 32}{\log_3 \sqrt{243}}$	p. positive integer
B. $\frac{2\log 6}{\log 12 + \log 3}$	q. negative integer
C. $\log_{1/4}\left(\frac{1}{16}\right)^{-2}$	r. rational but not integer
D. $\frac{\log_5 16 - \log_5 4}{\log_5 128}$	s. prime

निम्न Columns को सुमेलित कीजिए :

निम्नलिखित व्यजक सरल होने पर विघटित होगॉ

Column-I	**Column-II**
A. $\frac{\log_2 32}{\log_3 \sqrt{243}}$	p. धनात्मक पूर्णांक
B. $\frac{2\log 6}{\log 12+\log 3}$	q. ऋणात्मक पूर्णांक
C. $\log_{1/4}\left(\frac{1}{16}\right)^{-2}$	r. परिमेय लेकिन पूर्णांक न हो
D. $\frac{\log_5 16-\log_5 4}{\log_5 128}$	s. अभाज्य

(A) A - r; B - q; C - p,s; D - p

(B) A - r; B - q; C - p; D - p,s

(C) A - p,s; B - p; C - q; D - r

(D) A - q; B - r; C - p,s; D - p

Ans. C

Sol. (A) $\frac{\log_2 32}{\log_3 \sqrt{243}}=\frac{\log_2 2^5}{\log_3 \sqrt{3^5}}=\frac{5}{\log_3 3^{\frac{5}{2}}}=\frac{5}{\frac{5}{2}\times 1}=2$

(B) $\frac{2\log 6}{\log 12+\log 6}=\frac{2\log 6}{2\log 2+\log 3+\log 3}=\frac{2(\log 2+\log 3)}{2(\log 2+\log 3)}=1$

(C) $\log_{\frac{1}{4}}\left(\frac{1}{16}\right)^{-2}=\log_{2^{-2}} 16=\frac{1}{-2}\log_2 2^4=\frac{4}{-2}\log_2 2=-2$

(D) $\frac{\log_5 16-\log_5 4}{\log_5 128}=\frac{\log_5 \frac{16}{4}}{\log_5 2^7}=\frac{2\log_5 2}{7\log_5 2}=\frac{2}{7}$

Solve the following equations : [Take base 10 if not mentioned]

1. $\log_{x-1} 3 = 2$

Ans. $1+\sqrt{3}$

Sol. $\log_{x-1} 3 = 2$, $x-1 \neq 1$ and $x-1 > 0$

$\Rightarrow (x-1)^2 = 3$ $x \neq 2$ $x > 1$

$\Rightarrow x^2 - 1 - 2x = 0$

$D = b^2 - 4ac = 12$

$$\text{Roots} = \frac{-b+\sqrt{D}}{2a} = \frac{2 \pm 2\sqrt{3}}{2} = \frac{2(1\pm\sqrt{3})}{2} = 1 \pm \sqrt{3}$$

Since $1+\sqrt{3}$ satisfies the equation

2. $\log_4(2\log_3(1+\log_2(1+3\log_3 x))) = \frac{1}{2}$

Ans. 3

Sol. $\log_4(2\log_3(1+\log_2(1+3\log_3 x))) = \frac{1}{2} \Rightarrow \frac{1}{2}\log_2(2\log_3(1+\log_2(1+3\log_3 x))) = \frac{1}{2}$

$\Rightarrow 2\log_3(1+\log_2(1+3\log_3 x))) = 2^1 = 2 \Rightarrow (1+\log_2(1+3\log_3 4) = 3^1$

$\Rightarrow \log_2(1+3\log_3 x) = 3-1 \Rightarrow 1+3\log_3 x = 2^2$

$\Rightarrow 3\log_3 x = 3 \Rightarrow \log_3 x = 1$

$\Rightarrow x = 3^1 = 1$ $\therefore$ Final solution $x \in \{3\}$

3. $\log_3(1+\log_3(2x-7)) = 1$

Ans. 8

Sol. $\log_3(1+\log_3(2x-7) = 1 \Rightarrow 1+\log_3(2x-7) = 3 \Rightarrow (2x-7) = 3^2$

$\Rightarrow 2^x = 9+7 = 16 \Rightarrow x = 4$ $\therefore$ Final solution $x = \{4\}$

4. $\log_3(3^x - 8) = 2 - x$

Ans. 2

Sol. $\log_3(3^x-8) = 2-x \Rightarrow 3^{2-x} = 3^x - 8 \Rightarrow \frac{9}{3^x} = 3^x - 8$

Let $3^x = t$

$\frac{9}{t} = t - 8 \quad \Rightarrow \quad 9 = t^2 - 8t \quad \Rightarrow t^2 - 8t - 9 = 0$

$D = b^2 - 4ac = 64 + 36 = 100$

$t = 3^x = \frac{8 \pm 10}{2} = \frac{18}{2}, -\frac{2}{2} = 9, -1$ (negative value is not possible as $a^x > 0$)

$3^x = 9 \Rightarrow 3^x = 3^2 \Rightarrow x = 2$ $\therefore$ Final solution $x \in \{2\}$

5. $\frac{\log_2(9 - 2^x)}{3 - x} = 1$

Ans. 0

Sol. $\frac{\log_2(9 - 2^x)}{3 - x} = 1$

$\log_2(9 - 2^x) = 3 - x \; ; \; x \neq 3$

$9 - 2^x = 2^{3-x}$

$9 - 2^x = \frac{2^3}{2^x}$

put $2^x = t$

$(9 - t) = \frac{8}{t}$

$9t - t^2 - 8 = 0$

$t^2 - 9t + 18 = 0$

$t^2 - 8t - t + 8 = 0$

$t(t - 8)(t - 8) = 0$

$(t - 1)(t - 8) = 0$ but

$t = 1, t = 8$ $9 - 2^x > 0$

$2^x = 1, 2^x = 8$ $2^x < 9$

$x = 0, x = 3$ for $x = 3$

$\therefore x = 0$ $\therefore x \neq 3$

6. $\log_{5-x}(x^2 - 2x + 65) = 2$

Ans. -5

Sol. $\log_{5-x}(x^2 - 2x + 65) = 2 \Rightarrow (5 - x)^2 = x^2 - 2x + 65 \Rightarrow 25 + x^2 - 10x = x^2 - 2x + 65$

$\Rightarrow 25 - 65 = 8x \Rightarrow -40 = 8x \Rightarrow x = -5$

$\therefore$ Final solution $x \{-5\}$

7. $\log_3\left(\log_9 x+\frac{1}{2}+9^x\right)=2x$

Ans. 1/3

Sol. $\Rightarrow \log_3\left(\log_9 x+\frac{1}{2}+3^{2x}\right)=2x \qquad \Rightarrow -\frac{1}{2}=\frac{1}{2}\log_3 x$

$-1=\log_3 x \qquad \Rightarrow x=\frac{1}{3} \qquad \therefore$ Final solution $x \in \{\ \}$

8. $\log_3 (x+1)+\log_3 (x+3)=1$

Ans. 0

Sol. $\log_3(x+1)(x+3)=1 \qquad \Rightarrow 3^1=(x+1)(x+3) \qquad \Rightarrow 3=x^2+x+3x+3$

$3=x^2+4x+3 \qquad \Rightarrow 0=x^2+4x \qquad \Rightarrow 0=x(x+4)$

$\therefore x=-4, 0$ since -4 doesn't satisfy equation $\qquad \therefore$ Final solution $x \in \{0\}$

9. $\log_7(2^x-1)+\log_7(2^x-7)=1$

Ans. 3

Sol. $\log_7(2^x-1)+\log_7(2^x-7)=1$

Let $2^x=t$

$\log_7(t-1)(t-7)=1 \qquad \Rightarrow 7^1=t^2-7t-t+7 \quad \Rightarrow 0=t^2-8t \quad \Rightarrow t=0, 8$

$2^x=8 \quad \Rightarrow x=3$

$2^x=0$ impossible $\qquad \therefore$ Final solution $x\ \{3\}$

10. $\log 5+\log(x+10)-1=\log(21x-20)-\log(2x-1)$

Ans. 3/2, 10

Sol. $\log 5+\log(x+10)-1=\log(21x-20)-\log(2x-1) \qquad \Rightarrow \log(5+x+10)-\log_{10}10=\log$

$\Rightarrow \log\frac{5(x+10)}{10}=\log\frac{21x-20}{2x-1} \qquad \Rightarrow \log\frac{5(x+10)}{10}=\log\frac{21x-20}{2x-1}$

$\Rightarrow (2x-1)(x+10)=42x-40 \qquad \Rightarrow 2x^2-x+20x-10=42x-40$

$\Rightarrow 2x^2-19x-42x-10+40=0 \qquad \Rightarrow 2x^2-20x-3x+30=0$

$\Rightarrow (2x-3)(x-10)=0 \qquad \Rightarrow x=\frac{3}{2},10$

$\therefore$ Final solution $x \in \{3/2, 10\}$

11. $1-\log 5=\frac{1}{3}\left(\log\frac{1}{2}+\log x+\frac{1}{3}\log 5\right)$

Ans. $\frac{16}{\sqrt[3]{5}}$

Sol. $1-\log 5=\frac{1}{3}\left(\log\frac{1}{2}+\log x+\frac{1}{3}\log 5\right) \Rightarrow \log_{10}10-\log_{10}5=\frac{1}{3}\left[\log_{10}2^{-1}+\log_{10}x+\log_{10}5^{1/3}\right]$

$\Rightarrow \log_{10}\frac{10}{5}=\frac{1}{3}\left[\log_{10}2^{-1}+\log_{10}x+\log_{10}5^{1/3}\right] \Rightarrow \log_{10}2^3=\frac{1}{3}\log_{10}2^{-1}+\log_{10}x+\log_{10}5^{1/3}$

$\Rightarrow \log_{10}2^3=\log_{10}\left[\frac{x}{2}\times\sqrt[3]{5}\right] \Rightarrow x=\frac{16}{\sqrt[3]{5}}$

$\therefore$ Final solution $x\in\{\ \}$

12. $\log x-\frac{1}{2}\log\left(x-\frac{1}{2}\right)=\log\left(x+\frac{1}{2}\right)-\frac{1}{2}\log\left(x+\frac{1}{8}\right)$

Ans. 1

Sol. $\log x-\frac{1}{2}\log\left(x-\frac{1}{2}\right)=\log\left(x+\frac{1}{2}\right)-\frac{1}{2}\log\left(x+\frac{1}{8}\right) \Rightarrow \log x-\log\sqrt{x-\frac{1}{2}}=\log\left(x+\frac{1}{2}\right)-\log\sqrt{x+\frac{1}{8}}$

$\Rightarrow \log\frac{x}{\sqrt{x-\frac{1}{2}}}=\log\frac{x+\frac{1}{2}}{\sqrt{x+\frac{1}{8}}} \Rightarrow \left(\frac{x}{\sqrt{x-\frac{1}{2}}}\right)^2=\left(\frac{x+\frac{1}{2}}{\sqrt{x+\frac{1}{8}}}\right)^2 \Rightarrow \frac{x^2}{x-\frac{1}{2}}=\frac{x^2+\frac{1}{4}+x}{x+\frac{1}{8}}$

$\Rightarrow x^3+\frac{x^2}{8}=x^3+\frac{x}{4}+x^2-\frac{x^2}{2}-\frac{1}{8}-\frac{x}{2} \Rightarrow 8\left(8x^3+x^2\right)=8\left(8x^3+2x+8x^2-4x^2-1-4x\right)$

$\Rightarrow 8x^3+x^2=8x^3-2x+4x^2-1 \Rightarrow 0=3x^2-2x-1 \Rightarrow 0=3x^2-3x+x-1$

$\Rightarrow 0=3x(x-1)+1(x-1) \Rightarrow x=-\frac{1}{3},1$ $\therefore$ Final solution $x\in\{1\}$

13. $3^{\log_3\log\sqrt{x}}-\log x+\log^2 x-3=0$

Ans. 100

Sol. $3^{\log_3\left(\log\sqrt{x}\right)}-\log x+\log^2 x-3=0$

$\left(\log\sqrt{x}\right)^{\log_3 3} - \log x + \log^2 x - 3 = 0$

$\frac{1}{2}\log x - \log x + \log^2 x - 3 = 0$

$\log^2 x - \frac{1}{2}\log x - 3 = 0$

put $\log x = t$

$t^2 - \frac{t}{2} - 3 = 0$

$2t^2 - t - 6 = 0$

$2t^2 - 4t + 3t - 6 = 0$

$2t(t-2) + 3(t-2) = 0$

$(t-2)(2t+3) = 0$

$t = 2, t = -\frac{3}{2}$

for $t = 2$	for $t = -\frac{3}{2}$
$\log_{10} x = 2$	$\log_{10} x = -\frac{3}{2}$
$x = 10^2 = 100$	$x = \frac{1}{10\sqrt{10}}$

14. $(x-2)^{\log^2(x-2)+\log(x-2)^5-12} = 10^{2\log(x-2)}$

Ans. $2 + 10^{-7}, 3, 102$

Sol. $(x-2)^{\log^2(x-2)+\log(x-2)^5-12} = 10^{2\log(x-2)}$ (Let $\log(x-2) = t$)

$\Rightarrow \log(x-2)^{\log^2(x-2)+\log(x-2)^5-12} = \log 10^{2\log(x-2)}$

$\Rightarrow \log(x-2)\left(\log^2(x-2) + \log(x-2)^5 - 12\right) = 2\log(x-2)$; put $t = \log(x-2)$

$\Rightarrow t[t^2 + 5t - 12)] = 2t \quad \Rightarrow \quad t^3 + 5t^2 - 14t = 0 \quad \Rightarrow \quad t(t^2 + 5t - 14) = 0$

$\Rightarrow t(t+7)(t-2) = 0 \quad \Rightarrow \quad t = 0, -7, 2 \quad \Rightarrow \quad \log(x-2) = 0, -7, 2$

$\Rightarrow x = 2, 2 + 10^{-7}, 102$

15. $9^{\log_3(1-2x)} = 5x^2 - 5$

Ans. $-2 - \sqrt{10}$

Sol. $9^{\log_3(1-2x)} = 5x^2 - 5$

$\Rightarrow \quad 3^{2\log_3(1-2x)} = 5x^2 - 5$

$\Rightarrow \quad 3^{\log_3(1-2x)^2} = 5x^2 - 5$

$\Rightarrow \quad (1-2x)^2 = 5x^2 - 5$

$\Rightarrow \quad 1 + 4x^2 - 4x = 5x^2 - 5$

$\Rightarrow \quad 0 = 5x^2 - 4x^2 + 4x - 5 - 1$

$\Rightarrow \quad 0 = x^2 + 4x - 6$

$D = b^2 - 4ac$

$= 16 + 24 = 40, \sqrt{D} = 2\sqrt{10}$

$x = \dfrac{-b \pm \sqrt{D}}{2a} = -2 + \sqrt{10}, -2 - \sqrt{10}$

final : $x = -2 - \sqrt{10}$

16. $x^{1+\log x} = 10x$

Ans. 10^{-1}, 10

Sol. $x^{1+\log x} = 10x$

$(1 + \log x)\log x = (1 + \log x)$

$(1 + \log x)(\log x - 1) = 0$

$\log x = 1, \log x = -1$

$x = 10, \; x = \dfrac{1}{10}$

17. $x^{2\log x} = 10x^2$

Ans. $\sqrt{10^{1-\sqrt{3}}}, \sqrt{10^{1+\sqrt{3}}}$

Sol. $x^{2\log x} = 10x^2$

$\Rightarrow \quad \log_{10} x^2 \cdot \log x = \log_{10} 10 + \log_{10} x^2$

$\Rightarrow \quad 2\log_{10} x\left(\log_{10} x\right) = 1 + 2\log_{10} x$

$\Rightarrow \quad 2\log_{10}^2 x - 2\log_{10} x - 1 = 0 \qquad$ Let $\qquad \log_{10} x = t$

$\Rightarrow \quad 2t^2 - 2t - 1 = 0$

$D = b^2 - 4ac$

$= 4 + 8 = 12, \quad \sqrt{D} = 2\sqrt{3}$

$$t = \frac{2 \pm 2\sqrt{3}}{9} = \frac{2(1 \pm \sqrt{3})}{4} = \frac{1 \pm \sqrt{3}}{2}$$

Now, $\log_{10} x = t = \frac{1+\sqrt{3}}{2}, \frac{1-\sqrt{3}}{2}$

$$\Rightarrow \quad \log_{10} x = \frac{1 \pm \sqrt{3}}{2}$$

$$\Rightarrow \quad \log_{10} x^2 = 1 \pm \sqrt{3}$$

$$\Rightarrow \quad x^2 = 10^{1 \pm \sqrt{3}}$$

$$x = \sqrt{10^{1+\sqrt{3}}}, \sqrt{10^{1-\sqrt{3}}}$$

18. $x^{\frac{\log x + 5}{3}} = 10^{5 + \log x}$

Ans. $10^{-5}, 10^3$

Sol. $$\log x \cdot \frac{\log x + 5}{3} = \log_{10} 10^5 + \log_{10} x$$

$$\Rightarrow \quad \log x \cdot \frac{\log x + 5}{3} = 5 + \log_{10} x$$

$$\Rightarrow \quad \log^2 x + 5\log_{10} x = 15 + 3\log_{10} x$$

$$\Rightarrow \quad \log_{10}{}^2 x + 2\log_{10} x - 15 = 0$$

$$\Rightarrow \quad \log_{10} x\left(\log_{10} x + 5\right) - 3\left(\log_{10} x + 5\right) = 0$$

$$\Rightarrow \quad \left(\log_{10} x - 3\right)\left(\log_{10} x + 5\right) = 0$$

$$\Rightarrow \quad \log_{10} x = 3 \quad \log_{10} x = -5$$

$$\Rightarrow \quad x = 10^3 \quad x = 10^{-5}$$

19. $x^{\log_3 x} = 9$

Ans. $3^{-\sqrt{2}}, 3^{\sqrt{2}}$

Sol. $x^{\log_3 x} = 9$

$$\log_3 x \cdot \log_{10} x = \log_{10} 9$$

$$\frac{\log^2 x}{\log 3 \cdot \log 10} = \frac{\log 9}{\log 10}$$

$\log^2 x = 2\log^2 3$

$\log_3 x = 2^{\frac{1}{2}}, 2^{-\frac{1}{2}}$

$x = 3^{\sqrt{2}}, 3^{-\sqrt{2}}$

20. $\left(\sqrt{x}\right)^{\log_5 x - 1} = 5$

Ans. 1/5, 25

Sol. $\left(\sqrt{x}\right)^{\log_5 x - 1} = 5$

$\left(\log_5 x - 1\right)\log_{10}\sqrt{x} = \log_{10} 5$

$\log_5 x - 1 = \log_{\sqrt{x}} 5$

$\log_5 x - 1 = \dfrac{1}{\frac{1}{2}}\log_x 5 = 2\log_x 5$

$\left(\log_5 x - 1\right) = \dfrac{2}{\log_5 x}$

put $t = \log_5 x$
$(t-1)\cdot t = 2$
$t^2 - t - 2 = 0$
$t^2 - 2t + t - 2 = 0$
$t(t-2) + 1(t-2) = 0$
$(t-2)(t+1) = 0$
$t = 2, -1$
$\log_5 x = 2, \log_5 x = -1$
$x = 5^2, x = 5^{-1}$

$x = \left\{25, \dfrac{1}{5}\right\}$

21. $x^{\log x + 1} = 10^6$

Ans. 10^{-3}, 10^2

Sol. $x^{\log x + 1} = 10^6$

$\Rightarrow \log x(\log x + 1) = \log_{10} 10^6$

$\Rightarrow \log_{10}^2 x + \log_{10} x - 6 = 0$

$\Rightarrow \quad (\log_{10} x - 2)(\log_{10} x + 3) = 0$

$\Rightarrow \quad \log_{10} x = 2, \quad \log_{10} x = -3$

$\Rightarrow \quad x = 10^2 \quad , \qquad x = 10^{-3}$

22. $x^{\frac{\log x+7}{4}} = 10^{\log x+1}$

Ans. 10^{-4}, 10

Sol. $x^{\frac{\log x+7}{4}} = 10^{\log x+1}$

$\Rightarrow \left(\frac{\log x + 7}{4}\right)\log x = (\log x + 1)\cdot\log 10$

$\Rightarrow \quad \log_{10}^2 x + 3\log_{10} x - 4 = 0$

$\Rightarrow \quad \log_{10}^2 x + 4\log_{10} x - \log_{10} x - 4 = 0$

$\Rightarrow \quad (\log_{10} x + 4)(\log_{10} x - 1) = 0$

$\Rightarrow \quad x = 10^{-4}$ and $x = 10^1$

23. $x^{\log_{\sqrt{x}}(x-2)} = 9$

Ans. 5

Sol. $x^{\log_{\sqrt{x}}(x-2)} = 9$

$x^{2\log_x(x-2)} = 9$

$(x-2)^{2\log_2 x} = 9$

$(x-2)^2 = 9$

$x^2 + 4 - 4x - 9 = 0$

$x^2 - 4x - 5 = 0$

$x^2 - 5x + x - 5 = 0$

$x(x-5) + 1(x-5) = 0$

$(x-5)(x+1) = 0$

$x = 5, -1$

Final answer $x = 5$.

24. $\left(\frac{\log x}{2}\right)^{\log^2 x + \log x^2 - 2} = \log\sqrt{x}$

Ans. 10^{-3}, 10, 100

Sol. $\left(\frac{\log x}{2}\right)^{\log^2 x+\log^2 -2} = \log\sqrt{x}$

$\left(\frac{\log x}{2}\right)^{\log^2 x+2\log x-2} = \log\sqrt{x}$

$\left(\frac{\log x}{2}\right)^{\log^2 x+2\log x-2} = \frac{\log x}{2}$

taking log

$\left(\log^2 x+2\log x-2\right)\cdot\log\left(\frac{\log x}{2}\right) = \log\left(\frac{\log x}{2}\right)$

$\log\left(\frac{\log x}{2}\right)\left[\log^2 x+2\log x-3\right] = 0$

if $\log_{10}\left(\frac{\log x}{2}\right) = 0$

$\frac{\log x}{2} = 10^{\circ} = 1$

$\log_{10} x = 2$
$x = 10^2$
$x = 100$
if $\log^2 x + 2\log x - 3 = 0$
$\log^2 x + 3\log x - \log x - 3 = 0$
$(\log x + 3)(\log x - 1) = 0$
$\log x = -3, \log x = 1$
$x = 10^{-3}, x = 10$
$\therefore x = \{10, 100, 10^{-3}\}$

25. $3\sqrt{\log_2 x} - \log_2 8x + 1 = 0$

Ans. 2, 16

Sol. $3\sqrt{\log_2 x} - \log_2 8x + 1 = 0$

$\Rightarrow 3\sqrt{\log_2 x} - \left(\log_2 2^3 + \log_2 x\right) + 1 = 0$

$\Rightarrow 3\sqrt{\log_2 x} - 3 - \log_2 x + 1 = 0$

$\Rightarrow 3\sqrt{\log_2 x} - 2 - \log_2 x = 0$

$\Rightarrow \left(3\sqrt{\log_2 x}\right)^2 = \left(2 + \log_2 x\right)^2$

$\Rightarrow 9\log_2 x = 4 + 4\log_2 x + \log_2^2 x$

$\Rightarrow \quad 9\log_2 x - 4\log_2 x = 4 + \log_2{}^2 x$

$\Rightarrow \quad 0 = \log_2^2 x - 5\log_2 x + 4$

$\Rightarrow \quad 0 = \log_2^2 x - \log_2 x - 4\log_2 x + 4$

$\Rightarrow \quad 0 = \log_2 x(\log_2 x - 1) - 4(\log_2 x - 1)$

$\Rightarrow (\log_2 x - 1)(\log_2 x - 4) = 0$

$x = 2, 2^4$

$x = 2, 16$

26. $\log^2 x - 3\log x = \log(x^2) - 4$

Ans. $10, 10^4$

Sol. $\log^2 x - 3\log x = \log(x^2) - 4$

$\Rightarrow \log^2 x - 3\log x - 2\log x + 4 = 0$

$\Rightarrow \log_{10}{}^2 x - 5\log_{10} x + 4 = 0$

$\Rightarrow \log_{10}{}^2 x - \log_{10} x - 4\log_{10} x + 4 = 0$

$\Rightarrow (\log_{10} x - 4)(\log_{10} x - 1) = 0$

$\Rightarrow \log_{10} x = 4 \qquad \log_{10} x = 1$

$\Rightarrow x = 10^4$

and $x = 10$

27. $\log_{1/3} x - 3\sqrt{\log_{1/3} x} + 2 = 0$

Ans. $1/81, 1/3$

Sol. $\log_{1/3} x - 3\sqrt{\log_{1/3} x} + 2 = 0$

$\Rightarrow \quad (2 + \log_3 x)^2 = \left(3\sqrt{\log_3 x}\right)^2$

$\Rightarrow \quad 4 + \log_2^2 x + 4\log_3 x - 9\log_3 x = 0$

$\Rightarrow \quad \log_3^2 x + 5\log_3 x + 4 = 0$

$\Rightarrow \quad (\log_3 x + 4)(\log_3 x + 1) = 0$

$\Rightarrow \quad \log_3 x = -4 \quad \log_3 x = -1$

$\Rightarrow \quad 3^{-4} = x, \quad 3^{-1} = x$

$\Rightarrow \quad x = \frac{1}{81}, \frac{1}{3}$

28. $2(\log_x \sqrt{5})^2 - 3\log_x \sqrt{5} + 1 = 0$

Ans. $\sqrt{5}$; 5

Sol. $2\left(\log_x \sqrt{5}\right)^2 - 3\log_x \sqrt{5} + 1 = 0$

$$\Rightarrow \frac{2}{\left(\log_{\sqrt{5}} x\right)^2} - \frac{3}{\log_{\sqrt{5}} x} + 1 = 0$$

$$\Rightarrow \frac{2}{4\log_5^2 x} - \frac{3}{2\log_5 x} + 1 = 0$$

$$\Rightarrow 0 = 2\log_2^2 x - 3\log_5 x + 1$$

$$\Rightarrow 0 = 2\log_5 x\left(\log_5 x - 1\right) - 1\left(\log_5 x - 1\right)$$

$$\Rightarrow \log_5 x = \frac{1}{2}, \quad \log_5 x = 1$$

$$\Rightarrow x = \sqrt{5}, \quad x = 5$$

29. $\log_2^2 x + 2\log_2 \sqrt{x} - 2 = 0$

Ans. 1/4, 2

Sol. $\log_2^2 x + \frac{2}{2}\log_2 x - 2 = 0$

$$\Rightarrow \log_2^2 x + \log_2 x - 2 = 0$$

$$\Rightarrow \left(\log_2 x - 1\right)\left(\log_2 x + 2\right) = 0$$

$$\Rightarrow \log_2 x = 1 \quad \log_2 x = -2$$

$$\Rightarrow x = 2^1 \qquad x = 2^{-2}$$

$$\Rightarrow = 2 \qquad = \frac{1}{4}$$

30. $\left(a^{\log_b x}\right)^2 - 5x^{\log_b a} + 6 = 0$

Ans. $\left\{2^{\log_a b}, 3^{\log_a b}\right\}, a > 0, a \neq 1, b > 0, b \neq 1$

Sol. $\left(a^{\log_b x}\right)^2 - 5x^{\log_b a} + 6 = 0$

$\left(x^{\log_b a}\right)^2 - 5 \cdot x^{\log_b a} + 6 = 0$

Let $x^{\log_b a} = t$

$t^2 - 5t + 6 = 0$

$(t - 2)(t - 3) = 0$

$t = 2, 3$

for $t = 2$

$x^{\log_b a} = 2$

$x = 2^{\log_a b}$

for $t = 3$

$x^{\log_a b} = 3$

$x = 3^{\log_a b}$

$a > 0, a \neq 1, b \neq 1$

Solve the following equations : [Take base 10 if not mentioned]

1. $\log^2(100x) + \log^2(10x) = 14 + \log\left(\frac{1}{x}\right)$

Ans. $\sqrt{10^{-9}}, 10$

Sol. $\log^2(100x) + \log^2(10x) = 14 + \log\left(\frac{1}{x}\right)$

$(\log 100 + \log x)^2 + (\log 10 + \log x)^2 = 14 - \log x$

$(2 + \log x)^2 + (1 + \log x)^2 = 14 - \log x$

Let $\log_{10} x = t$

$(2 + t)^2 + (1 + t)^2 = 14 - t$

$4 + 4t + t^2 + 1 + 2t + t^2 = 14 - t$

$2t^2 + 6t + 5 + t - 14 = 0$

$2t^2 + 7t - 9 = 0$

$2t^2 + 9t - 2t - 9 = 0$

$t(2t + 9) - 1(2t + 9) = 0$

$(2t + 9)(t - 1) = 0$

$t = -\frac{9}{2}, 1$

for $t = -\frac{9}{2}$

$\log_{10} x = -\frac{9}{2}$

$x = 10^{-\frac{9}{2}}$

$x = \sqrt{10^{-9}}$

for $t = 1$

$\log^{10} x = 1$

$x = 10$

2. $\log_4(x+3) - \log_4(x-1) = 2 - \log_4 8$

Ans. 5

Sol. $\Rightarrow \log_4 \frac{x+3}{x-1} = 2 - \log_{2^2} 2^3$ $\qquad \frac{\log x}{\log_b a}$

$\Rightarrow \log_4 \frac{x+3}{x-1} = 2 - \frac{3}{2}$

$\Rightarrow \log_4 \frac{x+3}{x-1} = \frac{1}{2}$

$\Rightarrow 4^{1/2} = \frac{x+3}{x-1}$

$\Rightarrow 2(x-1) = (x+3)$

$\Rightarrow 2x - 2 = x + 3$

$\Rightarrow x = 5$

3. $2\log_3 \frac{x-3}{x-7} + 1 = \log_3 \frac{x-3}{x-1}$

Ans. –5

Sol. $2\log_3 \frac{x-3}{x-7} + 1 = \log_3 \frac{x-3}{x-1}$

$\log_3 \left(\frac{x-3}{x-7}\right)^2 + 1 = \log_3 \left(\frac{x-3}{x-1}\right)$

$\log_3 \left(\frac{x-3}{x-7}\right)^2 - \log_3 \left(\frac{x-3}{x-1}\right) = -1$

$\log_3 \frac{(x-3)^2}{(x-7)^2} \times \frac{x-1}{x-3} = -1$

$\frac{(x-3)(x-1)}{(x-7)^2} = 3^{-1}$

$\frac{x^2 - 4x + 3}{x^2 - 14x + 49} = \frac{1}{3}$

$3x^2 - 12x + 9 = x^2 + 14x - 49$

$\Rightarrow 2x^2 + 2x - 40 = 0$

$\Rightarrow x^2 + x - 20 = 0$

$\Rightarrow x(x+5) - 4(x+5) = 0$

$(x-4)\,(x+5) = 0$

$x = 4, -5$
for $x = 4$
since 4 can't satisfied the equation
$\therefore$ $x = -5$ is a solution

4. $2\log_4(4-x) = 4 - \log_2(-2-x)$

Ans. -4

Sol. $2\log_4(4-x) = 4 - \log_2(-2-x)$

$2\log_{2^2}(4-x) + \log_2(-2-x) = 4$

$\log_2(4-x)(-2-x) = 4$

$2^4 = x^2 - 2x - 8$

$0 = x^2 - 2x - 8 - 16$

$0=(x-6)(x+4)$

$\therefore \quad x = 6, -4$

final answer $x = -4$

5. $3 + 2\log_{x+1} 3 = 2\log_3(x+1)$

Ans. $-(3-\sqrt{3})/3, 8$

Sol. $3 + 2\log_{x+1} 3 = 2\log_3(x+1)$

Let $\log_3(x+1) = t$

$$\Rightarrow \quad 3 + \frac{2}{\log_3(x+1)} = 2\log_3(x+1)$$

$$\Rightarrow \quad = 3 + \frac{2}{t} = 2t$$

$$\Rightarrow \quad 3t + 2 = 2t^2$$

$$\Rightarrow \quad 0 = 2t^2 - 3t - 2$$

$$\Rightarrow \quad 0 = 2t^2 - 4t + t - 2$$

$$\Rightarrow \quad 0 = 2t(t-2) + 1(t-2)$$

$$\Rightarrow \quad 0 = (2t+1)(t-2) \qquad \Rightarrow t = 2, -\frac{1}{2}$$

$$\therefore \quad \log_3(x+1) = -\frac{1}{2} \qquad \text{or} \qquad \log_3 x + 1 = 2$$

$x + 1 = 3^{-1/2}$ $\qquad\qquad$ $x + 1 = 3^2$

$x = \frac{1}{\sqrt{3}} - 1$ $\qquad\qquad$ $x = 8$

6. $\log_x (9x^2).\log_3^2 x = 4$

Ans. 1/9, 3

Sol. $\log_x 9x^2 \cdot \log_3^2 x = 4$

$\left(\log_x 9 + \log_x x^2\right)\log_3^2 x = 4$

$\left(2\log_x 3 + 2\right)\cdot\log_3^2 x = 4$

$2\left(\frac{1}{\log_3 x} + 1\right)\cdot\log_3^2 x = 4$

Let $\log_3 x = t$

$\left(\frac{1}{t} + 1\right)\cdot t^2 = 2$

$(1 + t)t^2 = 2t$

$(1 + t)t^2 - 2t = 0$

$t\cdot((1 + t)t - 2) = 0$

$t\cdot(t^2 + t - 2) = 0$

$t\,(t^2 + 2t - t - 2) = 0$

$t(t\,(t + 2) - 1(t + 2)) = 0$

$t\,(t - 1)\,(t + 2) = 0$

$t = 0,\ 1,\ -2$

for $t = 0$

$\log_3 x = 0$

$x = 3^0 = 1$ (Impossible)

for $t = 1$

$\log_3 x = 1$

$x = 3$

for $t = -2$

$\log_3 x = -2$

$x = 3^{-2}$

$x = \frac{1}{9}$

7. $\log_{1/2}^2 (4x) + \log_2\left(\frac{x^2}{8}\right) = 8$

Ans. 2^{-7}, 2

Sol. $\log_{\frac{1}{2}}^{2}(4x)+\log_2\left(\frac{x^2}{8}\right)=8$

$\left(\log_{2^{-1}}(4x)\right)^2+\log_2\left(\frac{x^2}{8}\right)=8$

$\left(-\log_2 4-\log_2 x\right)^2+\left(\log_2 x^2-\log_2 8\right)=8$

$\left(-2-\log_2 x\right)^2+\left(2\log_2 x-3\right)=8$

$\left(2+\log_2 x\right)^2+2\log_2 x-3=8$

$\left(2+\log_2 x\right)^2+2\log_2 x-11=0$

let $\log_2 x = t$

$(2 + t)^2 + 2t - 11 = 0$

$4 + t^2 + 4t + 2t - 11 = 0$

$t^2 + 6t - 7 = 0$

$t^2 + 7t - t - 7 = 0$

$t(t + 7) - 1(t + 7) = 0$

$(t + 7)(t - 1) = 0$

$t = -7, 1$

for $t = -7$

$\log_2 x = -7$

$x = 2^{-7}$

for $t = 1$

$\log_2 x = 1$

$x = 2$

8. $\log_{0.5x} x^2 - 14\log_{16x} x^3 + 40\log_{4x}\sqrt{x} = 0$

Ans. $1/\sqrt{2}, 1, 4$

Sol. $\log_{0.5x} x^2 - 14\log_{16x} x^3 + 40\log_{4x}\sqrt{x} = 0$

$2\log_{0.5x} x - 42\log_{16x} x + \frac{40}{2}\cdot\log_{4x} x = 0$

$\frac{2}{\log_x 0.5x} - \frac{42}{\log_x 16x} + \frac{200}{\log_x 4x} = 0$

$2\cdot\left[\frac{\log_2 x}{\log_2 0.5 + \log x} - \frac{21\log_2 x}{\log_2 16 + \log_2 x} + \frac{10\log_2 x}{\log_2 4 + \log_2 x}\right] = 0$

$$\log_2 x\left[\frac{1}{\log_2 x-1}-\frac{21}{\log_2 x+4}+\frac{10}{\log_2 x+2}\right]=0$$

if $\log_2 x=0$

$x=1$

and if

$$\frac{1}{\log_2 x-1}-\frac{21}{\log_2 x+4}+\frac{10}{\log_2 x+2}=0$$

put $\log_2 x=t$

$$\frac{1}{(t-1)}-\frac{21}{(t+4)}+\frac{10}{(t+2)}=0$$

$(t+4)(t+2)-21(t-1)(t+2)+10(t-1)(t+4)=0$

$(t^2+6t+8)-21(t^2+t-2)+10(t^2+3t-4)=0$

$t^2-21t^2+10t^2+6t-21t+30t+8+42-40=0$

$-10t^2+15t+10=0$

$2t^2-3t-2=0$

$2t^2-4t+t-2=0$

$2t(t-2)+1(t-2)=0$

$(t-2)(2t+1)=0$

$$t=2,-\frac{1}{2}$$

$$\log_2 x=2,-\frac{1}{2}$$

$$x=2^2,2^{-\frac{1}{2}}$$

$$x=4,\frac{1}{\sqrt{2}}$$

9. $6-(1+4\cdot 9^{4-2\log_{\sqrt{3}}3})\cdot\log_7 x=\log_x 7, x\in Q$

Ans. 7

Sol. $6-\left(1+4\cdot 9^{4-2\log_{\sqrt{3}}3}\right)\cdot\log_7 x=\log_x 7, x\in Q$

$$6-\left(1+4\cdot 9^{4-2\times 2\log_3 3}\right)\cdot\log_7 x=\log_x 7$$

$$6-(1+4\times 1)\log_7 x=\log_x 7$$

$$6-5\cdot\log_7 x=\frac{1}{\log_7 x}$$

put $\log_7 x = t$

$6 - 5t = \frac{1}{t}$

$6t - 5t^2 = 1$

$5t^2 - 6t + 1 = 0$

$5t^2 - 5t - t + 1 = 0$

$5t(t-1) - 1(t-1) = 0$

$(t-1)(5t-1) = 0$

$t = 1, \frac{1}{5}$

for $t = \frac{1}{5}$

$\log_7 x = \frac{1}{5}$

$x = 7^{\frac{1}{5}}$ (Irrational)

for $t = 1$

$\log_7 x = t$

$x = 7$

10. $\log_3(4 \cdot 3^x - 1) = 2x + 1$

Ans. $-1, 0$

Sol. $\log_3\left(4.3^x - 1\right) = 2x + 1$

$3^{2x+1} = 4 \cdot 3^x - 1$

$3^{2x} \cdot 3^1 = 4 \cdot 3^x - 1$

Let $3^x = t$

$= 3t^2 - 4t - 1 = 0$ $\quad 3^x = \frac{1}{3}$ $\quad \Rightarrow x = -1$

$3t^2 - 3t - t + 1 = 0$

$(3t-1)(t-1) = 0$ $\quad 3^x = 1$ $\quad \Rightarrow x = 0$

$t = \frac{1}{3}, 1$

11. $\log_3(3^x - 6) = x - 1$

Ans. 2

Sol. $\log_3(3^x - 6) = x - 1$

$3^{x-1} = 3^x - 6$ $\quad 3^x = 9$

$\frac{3^x}{3} = 3^x - 6$ $\qquad$ $x = 22$

Put $t = 3^x$

$\frac{t}{3} = t - 6$

$t = 3t - 18$

$-2t = -18$

$t = 9$

$\Rightarrow 3^x = 9$

$x = 2$

12. $\log_3(4^x - 3) + \log_3(4^x - 1) = 1$

Ans. 1

Sol. $\log_3(4^x - 3) + \log_3(4^x - 1) = 1$

Let $4^x = t$

$\log_3(4^x - 3)(4^x - 1) = 1$

$3^1 = (t - 3)(t - 1)$ $\qquad$ $4^x = 4$

$3 = t^2 - 4t + 3$ $\qquad$ $x = 1$

$0 = t(t - 4)$

$t = 0, 4$

if $t = 0$, $x = \phi$

13. $\log_3(\log^2_{1/2} x - 3\log_{1/2} x + 5) = 2$

Ans. 1/16, 2

Sol. $\log_3\left(\log^2_{\frac{1}{2}} x - 3\log_{1/2} x + 5\right) = 2$

$\log^2_{2^{-1}} x - 3\log_{2^{-1}} x + 5 = 3^2$

Put $\log_{\frac{1}{2}} x = t$

$t^2 = 3t + 5 - 9 = 0$

$t^2 - 3t - 4 = 0$

$t = 4, -1$

for $t = 4$; for $t = -1$

$\log_{\frac{1}{2}} x = 4$; $\log_{\frac{1}{2}} x = -1$

$x = \left(\frac{1}{2}\right)^4$; $x = \left(\frac{1}{2}\right)^{-1}$

$x = \frac{1}{16}$; $x = 2$

14. $\log_5\left(\frac{2+x}{10}\right) = \log_5\left(\frac{2}{x+1}\right)$

Ans. 3

Sol. $\log_5 \frac{2+x}{10} = \log_5\left(\frac{2}{x+1}\right)$

$\log_5 \frac{2+x}{10} - \log_5 \frac{2}{x+1} = 0$

$\log_5 \frac{(2+x)}{10} \cdot \frac{(x+1)}{2} = 0$

$\frac{2x + x^2 + 2 + x}{20} = 5^0$

$x^2 + 3x + 2 = 20$

$x^2 + 3x - 18 = 0$

$(x + 6)(x - 3) = 0$

$x = 3, -6$

final $x = 3$

15. $1 + 2\log_{(x+2)} 5 = \log_5 (x + 2)$

Ans. –9/5, 23

Sol. $1 + 2\log_{(x+2)} 5 = \log_5 (x + 2)$

$1 + \frac{2}{\log_5 (x+2)} = \log_5 (x+2)$

put $t = \log_5 (x + 2)$

$1 + \frac{2}{t} = t$

$t + 2 = t^2$

$t^2 - t - 2 = 0$

$t(t - 2) + 1(t - 2) = 0$

$(t - 2)(t + 1) = 0$

$t = 2, -1$

for $t = 2$

$\log_5(x+2) = 2$

$x + 2 = 5^2$

$x + 2 = 25$

$x = 23$

for $t = -1$

$\log_5(x+2) = -1$

$x + 2 = 5^{-1}$

$$x = \frac{1}{5} - 2$$

$$x = -\frac{9}{5}$$

16. $\log_4(2^{4x}) = 2^{\log_2 4}$

Ans. 2

Sol. $\log_4 2^{4x} = 2^{\log_2 4}$

$\log_4 2^{4x} = 4$

$2^{4x} = 4^4$

$2^{4x} = (2^2)^4$

$4x = 8$

$x = 2$

17. $$\log_2\left(\frac{x}{4}\right) = \frac{15}{\log_2 \frac{x}{8} - 1}$$

Ans. 1/2, 128

Sol. $$\log_2 = \frac{15}{\log_2 \frac{x}{8} - 1}$$ Let $\log_2 x = t$

$$\Rightarrow \log_2 x - \log_2 4 = \frac{15}{\log_2 x - \log_2 8 - 1}$$

$$\Rightarrow \log_2 x - 2 = \frac{15}{\log_2 x - 3 - 1}$$

$$\log_2 x - 2 = \frac{15}{\log_2 x - 4}$$

$$(t-2) = \frac{15}{(t-4)}$$

$$(t-2)(t-4) = 15$$

$$t^2 - 2t - 4t + 8 = 15$$

$$t^2 - 6t + 8 - 15 = 0$$

$$t^2 - 6t - 7 = 0$$

$$t^2 - 7t + t - 7 = 0$$

$t(t-7) + 1(t-7) = 0$

$(t+1)(t-7) = 0$

$t = -1, 7$

$\log_2 x = -1$ or $\log_2 x = 7$

$x = \frac{1}{2}$ $\quad$ $x = 2^7$

18. $$\frac{1-2(\log x^2)^2}{\log x - 2(\log x)^2} = 1$$

Ans. $1/\sqrt{10}, \sqrt[3]{10}$

Sol. $$\frac{1-2\left(\log x^2\right)^2}{\log x - 2(\log x)^2} = 1$$

$$\frac{1-2\left(2\log x\right)^2}{\log x - 2\left(\log x\right)^2} = 1$$

put $\log_{10} x = t$

$$\frac{1-2\left(2t\right)^2}{t - 2t^2} = 1$$

$$\frac{1-8t^2}{t-2t^2} = 1$$

$1 - 8t^2 = t - 2t^2$

$-8t^2 + 2t^2 - t + 1 = 0$

$-6t^2 - t + 1 = 0$

$6t^2 + t - 1 = 0$

$6t^2 + 3t - 2t - 1 = 0$

$3t(2t+1)-1(2t+1)=0$

$(2t+1)(3t-1)=0$

$t=-\frac{1}{2},\frac{1}{3}$

for $t=-\frac{1}{2}$

$\log_{10} x=-\frac{1}{2}$

$x=10^{-\frac{1}{2}}$

$x=\frac{1}{\sqrt{10}}$

for $t=\frac{1}{3}$

$\log_{10} x=\frac{1}{3}$

$x=10^{\frac{1}{3}}$

19. $\log_2(4\cdot 3^x-6)-\log_2(9^x-6)=1$

Ans. 1

Sol. $\log_2(4.3^x-6)-\log_2(9^x-6)=1$ Let $3^x=t$

$\log_2(4.t-6)-\log_2(x^2-6)=1$

$4t-6=2\left(t^2-6\right)$

$0=2t^2-12-4t+6$

$0=2t^2-4t-6$

$0=t^2-2t-3$

$t^2-3t+t-3=0$

$\therefore$ $t(t-3)+1(t-3)=0$

$(t+1)(t-3)=0$

$\therefore$ $t=3,-1$

$3^x=3$

$x=1$

$3^x=-1$ Rejected

$\therefore$ Answer = 1

20. $\frac{1}{2}\log(5x-4)+\log\sqrt{x+1}=2+\log 0.18$

Ans. 8

Sol. $\frac{1}{2}\log(5x-4)+\log\sqrt{x+1}=2+\log\frac{0.18}{100}$

$\frac{1}{2}\log_{10}(5x-4)+\frac{1}{2}\log_{10}x+1=2+\log_{10}18-\log_{10}100$

$\Rightarrow \quad \frac{1}{2}\log_{10}(5x-4)(x-1)=\log_{10}18$

$5x^2-4x+5x-4=324$

$5x^2+x-328=0$

$D=b^2-4ac=1+6560=6561$

$\sqrt{D}=6561$

$x\equiv-\frac{82}{10},8$

$\therefore$ Final Ans. 8

21. $\log_4(2\cdot4^{x-2}-1)+4=2x$

Ans. 2

Sol. $\log_4\left(2\cdot4^{x-2}-1\right)+4=2x$

$\log_4\left(2\cdot4^{x-2}-1\right)=(2x-4)$

$2\cdot4^{x-2}-1=4^{2x-4}$

$2\cdot\frac{4^x}{4^2}-1=\frac{\left(4^x\right)^2}{4^4}$

$2\cdot\frac{4^x}{16}-1=\frac{\left(4^x\right)^2}{256}$

$\frac{4^x}{8}-1=\frac{\left(4^x\right)^2}{256}$

put $4^x=t$

$\frac{t}{8}-1=\frac{t^2}{256}$

$$\frac{(t-8)}{8}=\frac{t^2}{256}$$

$32t - 256 = t^2$

$t^2 - 32t + 256 = 0$

$(t - 16)^2 = 0$

$t = 16$

$4^x = 16$

$4^x = 4^2$

$x = 2$

22. $\log_x \sqrt{5} + \log_x (5x) - 2.25 = (\log_x \sqrt{5})^2$

Ans. $\sqrt[5]{5}, 5$

Sol. $\log_x \sqrt{5} + \log_x (5x) - 2.25 = \left(\log_x \sqrt{5}\right)^2$

$\log_x \sqrt{5} + \log_x 5 + 1 - 2.25 = \left(\log_x \sqrt{5}\right)^2$

$\frac{1}{2}\log_x 5 + \log_x 5 - 1.25 = \frac{1}{4}(\log_x 5)^2$

$\frac{3}{2}\log_x 5 - 1.25 = \frac{1}{4}(\log_x 5)^2$

put $\log_x 5 = t$

$\frac{3}{2}t - 1.25 = \frac{1}{4}t^2$

$6t - 5 = t^2$

$t^2 - 6t + 5 = 0$

$t^2 - 5t - t + 5 = 0$

$t(t - 5) - 1(t - 5) = 0$

$(t - 5)(t - 1) = 0$

$t = 1, 5$

for $t = 1$

$\log_x 5 = 1$

$x^1 = 5$

$x = 5$

for $t = 5$

$\log_x 5 = 5$

$x^5 = 5$

$x = 5^{\frac{1}{5}}$

23. $\log(\log x) + \log(\log x^4 - 3) = 0$

Ans. 10

Sol. $\log(\log x) + \log(\log x^4 - 3) = 0$

$\log(\log x) = -\log(\log x^4 - 3)$

$$\log(\log x) = \log\frac{1}{(\log x^4 - 3)}$$

$$\log x = \frac{1}{4\log x - 3}$$

put $\log_{10} x = t$

$$t = \frac{1}{(4t-3)}$$

$4t^2 - 3t - 1 = 0$

$4t^2 - 4t + t - 1 = 0$

$4t(t-1) + 1(t-1) = 0$

$(t-1)(4t+1) = 0$

$$t = 1, -\frac{1}{4}$$

for $t = 1$

$\log_{10} x = 1$

$x = 10$

for $t = -\frac{1}{4}$

$$\log_{10} x = -\frac{1}{4}(-\text{ve})$$

24. $\log_3 x - 2\log_{1/3} x = 6$

Ans. 9

Sol. $\log_3 x - 2\log_{1/3} x = 6$

$\log_3 x + 2\log_3 x = 6$

$\log_3 x + \log_3 x^2 = 6$

$x^3 = 3^6$

$x^3 = (3^2)^3$

$x = 9$

25. $$\frac{2\log x}{\log(5x-4)} = 1$$

Ans. 4

Sol. $\dfrac{2\log x}{\log(5x-4)} = 1$

$\Rightarrow$ $2\log x = \log(5x-4)$

$\log x^2 = \log(5x-4)$

$x^2 - 5x + 4 = 0$

$x^2 - 4x - x + 4 = 0$

$x(x-4) - 1(x-4) = 0$

$(x-1)(x-2) = 0$

$x = 1, 4$

Answer = 4

1. If $4^{\log_2 \ln x} = \ln x - (\ln x)^2 + 1$, then x =

यदि $4^{\log_2 \ln x} = \ln x - (\ln x)^2 + 1$, तो x =

(A) 2 (B) $\sqrt{e}$ (C) e (D) e^2

Ans. C

Sol. If $4^{\log_2 \ln x} = \ln x - (\ln x)^2 + 1$ when

$(\ln x)^{\log_2 4} = \ln x - (\ln x)^2 + 1$ $\ln x - 1 = 0$

$(\ln x)^2 = \ln x - (\ln x)^2 + 1 = 0$ $x = e$

$2(\ln x)^2 - \ln x - 1 = 0$ when

$2(\ln x)^2 - 2(\ln x) + \ln x - 1 = 0$ $2\ln x + 1 = 0$

$2\ln x(\ln x - 1) + 1(\ln x - 1) = 0$ $\ln x = -\frac{1}{2}$

$(\ln x - 1)(2\ln x + 1) = 0$ $x = \frac{1}{\sqrt{e}}$

2. a = log 12, b = log 21, c = log 11 and d = log 22 then $\log\left(\frac{1}{7}\right)$ can be expressed in this form P(a – b) + Q(c – d) where P and Q are integers then the value of (7P – Q) equals :

यदि a = log 12, b = log 21, c = log 11 तथा d = log 22 तो $\log\left(\frac{1}{7}\right)$ को यदि P(a – b) + Q(c – d) के रूप में व्यक्त किया जाए जहां पर P व Q पूर्णांक है तो (7P – Q) का मान होगा :

(A) 5 (B) 9 (C) 13 (D) 15

Ans. A

Sol. $\log\left(\frac{1}{7}\right) = P\log\left(\frac{12}{21}\right) + Q\log\left(\frac{11}{22}\right) \Rightarrow \log\left(\frac{1}{7}\right) = P\log\left(\frac{4}{7}\right) + Q\log\left(\frac{1}{2}\right)$

$\log\left(\frac{1}{7}\right) = P\log 4 - P\log 7 - Q\log 2 \Rightarrow \log\left(\frac{1}{7}\right) = (2P - Q)\log 2 - P\log 7$

$P = 1;\ 2P - Q = 0 \Rightarrow Q = 2$

$\Rightarrow$ $7P - Q = 5$ Ans.

3. Solution set of the equation $3^{2x^2} - 2\cdot 3^{x^2+x+6} + 3^{2(x+6)} = 0$ is :

समीकरण $3^{2x^2} - 2\cdot 3^{x^2+x+6} + 3^{2(x+6)} = 0$ के हलो का समुच्चय है :

(A) $\{-3, 2\}$ (B) $\{6, -1\}$ (C) $\{-2, 3\}$ (D) $\{1, -6\}$

Ans. C

Sol. $\left(3^{x^2} - 3^{(x+6)}\right)^2 = 0 \Rightarrow x^2 = x + 6 \Rightarrow (x^2 - x - 6) = 0 \Rightarrow (x-3)(x+2) = 0 \Rightarrow x = 3$ or $x = -2$

4. Let $u = (\log_2 x)^2 - 6\log_2 x + 12$ where x is a real number. Then the equation $x^u = 256$ has :

(A) no solution for x (B) exactly one solution for x

(C) exactly two distinct solutions for x (D) exactly three distinct solutions for x

यदि $u = (\log_2 x)^2 - 6\log_2 x + 12$ जहाँ x एक वास्तविक संख्या है तो समीकरण $x^u = 256$ का :

(A) x के लिये कोई हल नहीं (B) x के लिये सिर्फ एक हल

(C) x के लिये सिर्फ दो हल (D) x के लिये सिर्फ तीन भिन्न भिन्न हल

Ans. B

Sol. Let $u = (\log_2 x)^2 - 6\log_2 x + 12$

let $\log_2 x = t$

$x = 2^t$

$u = (t^2 - 6t + 12)$

And $x^4 = 256$

$(2^t)^4 = 2^8$

$2^{4t} = 2^8$

$t\cdot(t^2 - 6t + 12) = 8$

$t^3 - 6t^2 + 12t - 8 = 0$

$t^3 - 2^3 - 6t(t-2) = 0$

$(t-2)^3 = 0$ $\because x = 2^t$

$t = 2$ $x = 2^2$

$x = 4$

5. The sum of the roots of the equation, $x + 1 - 2\log_2(3 + 2^x) + 2\log_4(10 - 2^{-x}) = 0$, is : [JEE Main Aug 2021]

समीकरण $x + 1 - 2\log_2(3 + 2^x) + 2\log_4(10 - 2^{-x}) = 0$ के मूलों का योग है :

(A) $\log_2 13$ (B) $\log_2 11$ (C) $\log_2 12$ (D) $\log_2 14$

Ans. B

Sol. $x + 1 - 2\log_2(3 + 2^x) + 2\log_4(10 - 2^{-x}) = 10$

$x+1=\log_2(3+2^x)^2-\log_2(10-2^{-x})$

$$x+1=\log_2\left(\frac{(3+2^x)^2}{10-2^{-x}}\right)$$

$$2^{x+1}=\frac{9+2^{2x}+6\times 2^x}{10-2^{-x}}$$

$$2\times 2^x=\frac{9+2^{2x}+6\times 2^x}{10-\frac{1}{2^x}}$$

$2^x=t$

$$2t=\frac{9+t^2+6t}{10-\frac{1}{t}}$$

$$2t\left(10-\frac{1}{t}\right)=9+t^2+6t$$

$t^2-14t+11=0 \Rightarrow t_1, t_2$

$t_1t_2=11$

$2^{x_1}\times 2^{x_2}=11$

$2^{x_1+x_2}=11$

$x_1+x_2=\log_2 11$

6. If the equation $2^x+4^y=2^y+4^x$ is solved for y in terms of x where $x<0$, then the sum of the solutions is :

यदि समीकरण $2^x+4^y=2^y+4^x$ को y के लिये x के पदों के रूप में हल किया जाता है $x<0$ ता हलो का योग :

(A) $x\log_2(1-2^x)$ (B) $x+\log_2(1-2^x)$ (C) $\log_2(1-2^x)$ (D) $x\log_2(2^x+1)$

Ans. B

Sol. $2^{2y}-2^y+2^x(1-2^x)=0$

put $2^y=t$

$t^2-t+2^x(1-2^x)=0$ where $t_1=2^{y_1}$ and $t_2=2^{y_2}$

$t_1t_2=2^x(1-2^x)$

$2^{y_1+y_2}=2^x(1-2^x)$

$y_1+y_2=x+\log_2(1-2^x)$ Ans.

7. The set of all real numbers x for which $x^2-|x+2|+x>0$, is :

x के सभी वास्तविक मानों का समुच्चय जिनके लिये $x^2 - |x + 2| + x > 0$ है :

(A) $(-\infty, -2) \cup (2, \infty)$ (B) $(-\infty, -\sqrt{2}) \cup (\sqrt{2}, \infty)$

(C) $[-2, -\sqrt{2}] \cup (\sqrt{2}, \infty)$ (D) $(\sqrt{2}, \infty)$

Ans. B

Sol. $x^2 - |x + 2| + x > 0$

Case I : $x < -2$

$x^2 + x + 2 + x > 0$

$x^2 + 2x + 2 > 0$

$D < 0, x \in R$

$\therefore x < -2$

Case II : $x \geq -2$

$x^2 - x - 2 + x > 0$

$x^2 - 2 > 0$

$(x - \sqrt{2})(x + \sqrt{2}) > 0$

$x < -\sqrt{2}, x > \sqrt{2} \Rightarrow x \in (-\infty, -\sqrt{2}) \cup (\sqrt{2}, +\infty)$

$\therefore$ Case I $\cup$ Case II

$x \in (-\infty, -\sqrt{2}) \cup (\sqrt{2}, +\infty)$

8. Sum of all the roots of the equation $x^2 - 2x + |x - 1| - 5 = 0$ is :

समीकरण $x^2 - 2x + |x - 1| - 5 = 0$ के सभी मूलों का योग होगा :

(A) 0 (B) 2 (C) 1 (D) 5

Ans. B

Sol. $x^2 - 2x + |x - 1| - 5 = 0$

Case I : $x \geq 1$

$x^2 - 2x + x - 1 - 5 = 0$

$x^2 - x - 6 = 0$

$x^2 - 3x + 2x - 6 = 0$

$(x - 3)(x + 2) = 0$

$x = -2, 3$ (–2 rejected)

Case II : $x < 1$

$x^2 - 2x - x + 1 - 5 = 0$

$x^2 - 3x - 4 = 0$

$x^2 - 4x + x - 4 = 0$

$x(x - 4) + 1(x - 4) = 0$

$(x - 4)(x + 1) = 0$

$x = 4, -1$ (4 rejected)

Sum of all roots = 3 – 1 = 2

9. The complete set of real 'x' satisfying $\big|\,|x-1|-1\,\big| \le 1$ is :

'x' का सम्पूर्ण समुच्चय जो $\big|\,|x-1|-1\,\big| \le 1$ का सन्तुष्ट करता है :

(A) [0, 2] (B) [– 1, 3] (C) [– 1, 1] (D) [1, 3]

Ans. B

Sol. $\big||x-1|-1\big| \le 1$

$-1 \le |x-1|-1 \le 1$

$0 \le |x-1| \le 2$

when

$|x-1| \ge 0$

when $x < 1$

$-x + 1 \ge 0$

$x \le 1 \quad \Rightarrow x < 1$

and when $x \ge 1$

$x - 1 \ge 0$

$x \ge 1 \quad \Rightarrow x \ge 1$

$x \in (-\infty, \infty)$

And

$|x-1| \le 2$

when $x - 1 < 0$

$x < 1$

$\therefore\ -x + 1 \le 2$

$-x \le 1$

$x \ge 1$

when $x - 1 > 0$

$x > 1$

$x - 1 \le 2$

$x \le 3$

$\because\ x \ge -1, x \le 3$

$x \in [-1, 3]$

10. The number of the real roots of the equation $(x+1)^2 + |x-5| = \frac{27}{4}$ is _____.

समीकरण $(x+1)^2 + |x-5| = \frac{27}{4}$ के वास्तविक मूलों की संख्या है _____

(A) 4 (B) 3 (C) 1 (D) 2

Ans. D

Sol. C – I $x \geq 5$

$(x+1)^2 + x - 5 = \frac{27}{4}$

$x^2 + 3x - \frac{43}{4} = 0$ (Both roots < 5)

$\Rightarrow x \in \phi$

C – II $x < 5$

$(x+1)^2 - (x-5) = \frac{27}{4}$

$x^2 + x - \frac{3}{4} = 0$ (as both roots < 5)

$\Rightarrow$ 2 real roots

11. The number of real solutions of the equation, $x^2 - |x| - 12 = 0$ is :

समीकरण, $x^2 - |x| - 12 = 0$ के वास्तविक हलों की संख्या है :

(A) 4 (B) 3 (C) 1 (D) 2

Ans. D

Sol. $x^2 - |x| - 12 = 0$

$t^2 - t - 12 = 0$

$(t-4)(t+3) = 0 = t = 4, -3$

$|x| = 4$, $|x| = -3$ (rejected)

$x = \pm 4$

12. Let $S = \{x \in R: x \geq 0 \text{ and } 2|\sqrt{x} - 3| + \sqrt{x}(\sqrt{x} - 6) + 6 = 0\}$. Then S : [JEE(Main)-2018]

(A) contains exactly two elements (B) contains exactly four elements

(C) is an empty set (D) contains exactly one element

माना $S = \{x \in R: x \geq 0$ तथा $2|\sqrt{x} - 3| + \sqrt{x}(\sqrt{x} - 6) + 6 = 0\}$ तो S [JEE(Main)-2018]

(A) में मात्र दो अवयव हैं (B) में मात्र चार अवयव हैं

(C) एक रिक्त समुच्चय हैं (D) में मात्र एक ही अवयव हैं

Ans. A

Sol. $S = \{x \in R: x \geq 0 \text{ and } 2|\sqrt{x} - 3| + \sqrt{x}(\sqrt{x} - 6) + 6 = 0\}$

$2\left|\sqrt{x}3\right| + x - 6\sqrt{x} + 6 = 0$

when $\sqrt{x} < 3 \Rightarrow x < 9$

$2\left(3-\sqrt{x}\right)+x-6\sqrt{x}+6=0$

$6-2\sqrt{x}+x-6\sqrt{x}+6=0$

$12 + x = 8\sqrt{x}$

$x^2 + 144 + 24x = 64x$

$x^2 - 40x + 144 = 0$

$x^2 - 36x - 4x + 144 = 0$

$x(x - 36) - 4 (x - 36x) = 0$

$x = 4, 36$ (36 rejected)

when $\text{whe}\sqrt{x} \geq 3 \Rightarrow x \geq 9$

$2\left(\sqrt{x}-3\right)+x-6\sqrt{x}+6=0$

$2\sqrt{x}-6+x-6\sqrt{x}+6=0$

$x-\sqrt{4}x=0$

$x=\sqrt{4}x$

$x^2 = 16x$

$x = 0, 16$ (0 rejected)

$\therefore$ Contains exactly two elements.

13. If a and b are the roots of the equation $(\log_2 x)^2 + 4(\log_2 x) - 1 = 0$ then the value of $\log_b a + \log_a b$ equals :

यदि समीकरण $(\log_2 x)^2 + 4(\log_2 x) - 1 = 0$ के मूल α तथा β है तो $\log_\beta \alpha + \log_\alpha \beta$ का मान क्या होगा :

(A) 18 (B) -16 (C) 14 (D) -18

Ans. D

Sol. $\log_2\alpha + \log_2\beta = -4; \quad \log_2\alpha \cdot \log_2\beta = -1$

now $\log_\beta\alpha + \log_\alpha\beta = \dfrac{\log_2\alpha}{\log_2\beta}+\dfrac{\log_2\beta}{\log_2\alpha}=\dfrac{(\log_2\alpha)^2+(\log_2\beta)^2}{\log_2\alpha\cdot\log_2\beta}$

$= -[(\log_2\alpha + \log_2\beta)^2 - 2\log_2\alpha \cdot \log_2\beta]$

$= -[16 + 2] = -18$ Ans.

ONE OR MORE THAN ONE CORRECT TYPE

14. The equation $\dfrac{\log_8\left(\dfrac{8}{x^2}\right)}{(\log_8 x)^2}=3$ has :

(A) no integral solution (B) one natural solution (C) two real solutions (D) one irrational solution

समीकरण $\dfrac{\log_8\left(\dfrac{8}{x^2}\right)}{(\log_8 x)^2}=3$ का :

(A) कोई पूर्णांक हल नहीं (B) एक प्राकृतिक हल (C) दो वास्तविक हल (D) एक अपरिमेय हल

Ans. BC

Sol. $\dfrac{1-2y}{y^2}=3 \qquad (\log_8 x = y)$

$3y^2+2y-1=0 \quad \Rightarrow \quad 3y^2+3y-y-1=0 \quad \Rightarrow \quad 3y(y+1)-1(y+1)=0 \Rightarrow y=\dfrac{1}{3},-1$

$\Rightarrow \log_8 x=\dfrac{1}{3}, \qquad \log_8 x=-1$

$x=2 \qquad\qquad x=\dfrac{1}{8}$

15. In which of the following case(s) the real number 'm' is greater than the real number 'n'?

निम्नलिखित विकल्पों में से किन विकल्पो में वास्तविक संख्या 'm' वास्तविक संख्या 'n' से बड़ी होगी ?

(A) $m=(\log_2 5)^2$ and $n=\log_2 20$ (B) $m=\log_{10}2$ and $n=\log_{10}\sqrt[3]{10}$

(C) $m=\log_{10}5\cdot\log_{10}20+(\log_{10}2)^2$ and $n=1$ (D) $m=\log_{\frac{1}{2}}\left(\dfrac{1}{3}\right)$ and $n=\log_{\frac{1}{3}}\left(\dfrac{1}{2}\right)$

Ans. AD

Sol. (A) $m-n=(\log_2 5)^2-[\log_2 5+2]$

let $\log_2 5=x$

$=x^2-x-2$

$=(x-2)(x+1)$

$=(\log_2 5-2)(\log_2 5+1)>0$

hence $m>n \Rightarrow$ (A) is Ans.

(B) $m=\log_{10}2=0.3010;\ n=\dfrac{1}{3}=0.333.....;$ hence $n>m$

(C) $m=(1-\log_{10}2)(1+\log_{10}2)$

$=1-(\log_{10}2)^2$ and $n=1$

$\therefore \quad m-n=-(\log_{10}2)^2<0$

$\therefore \quad m < n$

(D) $m = \log_2 3; \quad n = \log_3 2 \quad \Rightarrow \quad m > n$ (D) is Ans.

16. Let $(\log_2 x)^2 - 4\log_2 x - m^2 - 2m - 13 = 0$ be an equation in x and $m \in R$, then which of the following must be correct?

(A) For any $m \in R$, the equation has two distinct solutions

(B) The product of the solutions of the equation does not depend on m

(C) One of the solutions of the equation is less than 1 while the other is greater than 1 for $\forall\ m \in R$

(D) The minimum value of the larger solution is 2^6 and maximum value of the smaller solution is 2^{-2}

माना $(\log_2 x)^2 - 4\log_2 x - m^2 - 2m - 13 = 0$ एक समीकरण है तथा $m \in R$ है तो निम्न में से कौन सत्य है ?

(A) किसी $m \in R$ के लिये समीकरण के दो भिन्न हल होगें

(B) समीकरण के हलो का गुणन m पर निर्भर नही करता

(C) समीकरण का एक हल एक से छोटा है जबकि अन्य हल एक से बड़ा है, $\forall\ m \in R$ के लिये

(D) बड़े हल का न्यूनतम मान 2^6 है तथा छोटे हल का अधिकतम मान 2^{-2} है

Ans. ABCD

Sol. $(\log_2 x)^2 - 4\log_2 x - m^2 - 2m - 13 = 0$

let $\log_2 x = t,\ x > 0$

(A) $t^2 - 4t - (m^2 + 2m + 13) = 0$

$D = 16 + 4(m^2 + 2m + 13)$

$D = 16 + 4(m^2 + 2m + 13)$

$\quad +ve \quad +ve$

$D \geq 0$

$\therefore$ for any $m \in R$, the equation has two distinct solution.

$$\Rightarrow t^2 - 4t - m^2 - 2m - -13 = 0 \begin{cases} \log_2 x_1 \\ \log_2 x_2 \end{cases}$$

$\text{Sum} = \log_2 x_1 + \log_2 x_2 = 4$

$\log_2 x_1 x_2 = 4$

$x_1 x_2 = 2^4 = 16$

it does not depend on m.

$\Rightarrow f(1) < 0$

$(\log_2 1)^2 = 4(\log_2 1) - m^2 - 2m - 13 < 0$

$0 - 0 - m^2 - 2m - 13 < 0$

$m^2 + 2 + 13 > 0$

$\because\ (D < 0)$

$m \in R$

$\Rightarrow (\log_2 x)^2 - 4(\log_2 x) - m^2 - 2m - 13 = 0$

$$\log_2 x_1 = \frac{4 \pm \sqrt{16 + 4\left(m^2 + 2m + 13\right)}}{2}$$

$$= \frac{4 \pm 2\sqrt{4 + m^2 + 2m + 13}}{2}$$

$$\log_2 x_1 = 2 \pm \sqrt{m^2 + 2m + 17}$$

$$x_1 = 2^{2 \pm \sqrt{m^2 + 2m + 17}}$$

for $m^2 + 2m + 17$

$$\text{Minimum value} = \frac{-D}{4a} = \frac{4ac - b^2}{4a} = \frac{68 - 4}{4} = \frac{64}{4} = 16$$

$$\therefore x_1 = 2^{2 \pm \sqrt{16}} = 2^{2 \pm 4}$$

$x_1 =$ 2^6 (maximum) or 2^{-2} (minimum)

17. If $1 + \log_5(x^2 + 1) \geq \log_5(ax^2 + 4x + a)$, $\forall x \in R$ then 'a' can be equal to :

यदि $1 + \log_5(x^2 + 1) \geq \log_5(ax^2 + 4x + a)$, $\forall x \in R$ है, तो 'a' का मान हो सकता है :

(A) 3 (B) $\frac{5}{2}$ (C) 2 (D) $\frac{3}{2}$

Ans. AB

Sol. $1 + \log_5(x^2 + 1) \geq \log_5(a^2 + 4x + a)$, $\forall x \in R$

$\log_5 5(x^2 + 1) - \log_5 (ax^2 + 4x + a) \geq 0$

$$\log_5 \left(\frac{ax^2 + 4x + a}{5\left(x^2 + 1\right)} \right) \leq 0$$

$$\frac{ax^2 + 4a + a}{5x^2 + 5} \leq 1$$

$$\frac{ax^2 + 4a + a}{5x^2 + 5} - 1 \leq 0$$

$ax^2 + 4x + a - 5x^2 - 5 \leq 0$ $\left\{ \because 5x^2 + 5 > 0 \right.$

$(a - 5)x^2 + 4x + (a - 5) \leq 0$

$\Rightarrow a - 5 < 0$ (for purely –ve)

$a < 5$

$D \leq 0$

$16 - 4(a - 5)^2 \leq 0$

$4 - (a - 5)^2 \leq 0$

$(2 - a + 5)(2 + a - 5) \leq 0$

$(7 - a)(a - 3) \leq 0$

$(a-3)(a-7) \le 0$
$a \le 3, a \le 7$
$\Rightarrow a \le 3$

Paragraph for question nos. 18 to 20

Let A denotes the sum of the roots of the equation $\frac{1}{5-4\log_4 x}+\frac{4}{1+\log_4 x}=3$.

B denotes the value of the product of m and n, if $2^m = 3$ and $3^n = 4$.

C denotes the sum of the integral roots of the equation $\log_{3x}\left(\frac{3}{x}\right)+(\log_3 x)^2 = 1$.

प्रश्न 18 से 20 के लिये अनुच्छेद

माना कि A व्यक्त होगा समीकरण $\frac{1}{5-4\log_4 x}+\frac{4}{1+\log_4 x}=3$ के मूलो के योग से।

B व्यक्त होगा m व n के गुणनफल से जहाँ $2^m = 3$ व $3^n = 4$.

C व्यक्त होगा समीकरण $\log_{3x}\left(\frac{3}{x}\right)+(\log_3 x)^2 = 1$ के पूर्णांक मूलो से

18. The value of A + B equals :

A + B का मान होगा :

(A) 10 (B) 6 (C) 8 (D) 4

Ans. C

19. The value of B + C equals :

B + C का मान होगा :

(A) 6 (B) 2 (C) 4 (D) 8

Ans. A

20. The value of A + C ÷ B equals :

A + C ÷ B का मान होगा :

(A) 5 (B) 8 (C) 7 (D) 4

Ans. B

Sol. (A) We have

$$\frac{1}{5-4\log_4 x}+\frac{4}{1+\log_4 x}=3$$

Put $\log_4 x = y$

$$\Rightarrow \quad \frac{1}{5-4y}+\frac{4}{1+y}=3$$

$\Rightarrow \quad (1 + y) + 4(5 - 4y) = 3\ (5 - 4y)\ (1 + y)$

$\Rightarrow \quad 2y^2 - 3y + 1 = 0$

$$\therefore \quad y = 1,\ \frac{1}{2}$$

$$\Rightarrow \quad \log_4 x = 1 \ \&\ \log_4 x = \frac{1}{2}$$

$\therefore \quad x = 4 \ \&\ x = 2$

$\Rightarrow \quad$ sum of the roots of the equation $= 4 + 2 = 6$

(B) We have $2^m = 3$ and $3^n = 4$

$$\Rightarrow \quad m = \frac{\log 3}{\log 2} \text{ and } n = \frac{\log 4}{\log 3}$$

$$\therefore \quad mn = \frac{\log 3}{\log 2} \cdot \frac{\log 4}{\log 3} = \log_2 4 \ = 2$$

$\therefore \quad mn = 2$

(C) Now, we have clearly $0 < x \neq \frac{1}{3}$

$$\log_{3x} 3 - \log_{3x} x + (\log_3 x)^2 = 1$$

$$\Rightarrow \quad \frac{1}{\log_3 (3x)} - \frac{1}{\log_x (3x)} + (\log_3 x)^2 = 1$$

$$\Rightarrow \quad \frac{1}{1+\log_3 x} - \frac{1}{\log_x 3 + 1} + (\log_3 x)^2 = 1$$

Put $\log_3 x = y$, we get

$$\Rightarrow \quad \frac{1}{y+1} - \frac{1}{\frac{1}{y}+1} + y^2 = 1 \quad \Rightarrow \quad \frac{(1-y)}{y+1} + y^2 = 1 \quad \Rightarrow \quad (1 - y) + y^2(y + 1) = y + 1$$

$\Rightarrow \quad 1 - y + y^3 + y^2 - y - 1 = 0 \quad \Rightarrow \quad y^3 + y^2 - 2y = 0$

$\therefore \quad y(y^2 + y - 2) = 0 \quad$ or $\quad y(y + 2)(y - 1) = 0$

$\therefore \quad y = 0, -2, 1$

But $\log_3 x = y$ gives $n = 1, \frac{1}{9}, 3$

$\therefore \quad$ Sum of integral roots $= 1 + 3 = 4$

$A + B = 6 + 2 = 8$

$B + C = 2 + 4 = 6$

$A + C \div B = 6 + 4 \div 2 = 6 + 2 = 8$

21. Establish tricotomy in each of this following pairs of numbers :

प्रत्येक विकल्पों में >, < या = में से जो भी सम्बन्ध बनता है उसे स्थापित कीजिए :

(i) $3^{\log_{27} 3}$ and $2^{\log_4 2}$ (ii) $\log_4 5$ and $\log_{1/16}(1/25)$

(iii) 4 and $\log_3 10 + \log_{10} 81$ (iv) $\log_{1/5}(1/7)$ and $\log_{1/7}(1/5)$

Ans. (i) (Hint : >); (ii) (Hint : =); (iii) (Hint : <); (ii) (Hint : >);

Sol. (i) Let $a = 3^{\log_{27} 3} = 3^{\frac{1}{3}\cdot\log_3 3} = 3^{\frac{1}{3}} = \sqrt[3]{3}$

$b = 2^{\log_4 2} = 2^{\frac{1}{2}\log_2 2} = 2^{\frac{1}{2}} = \sqrt[2]{2}$

Now $a = \left(3^2\right)^{\frac{1}{6}} = 9^{\frac{1}{6}}$

$b = \left(2^2\right)^{\frac{1}{6}} = 8^{\frac{1}{6}}$ $\therefore\ a > b$

(ii) Let $a = \log_4 5$

$b = \log_{\frac{1}{16}}\left(\frac{1}{25}\right)$

$b = \log_{4^{-2}} 5^{-2} = \frac{-2}{-2}\log_4 5 = \log_4 5$

$\Rightarrow a = b$

(iii) Let $a = 4$,

$b = \log_3 10 + \log_{10} 81$ $b \geq 2 \times 2$

$b = \log_3 10 + \log_{10} 3^4$ $b \geq 4$

$b = \log_3 10 + 4\cdot\log_{10} 3$ $\log_3 10 + \frac{4}{\log_3 10} > 4$

$b = \log_3 10 + \frac{4}{\log_3 10}$ $b > 4$

$\because$ AM $\geq$ GM

$\frac{\log_3 10 + \frac{4}{\log_3 10}}{2} \geq \sqrt{\log_3 10 \times \frac{4}{\log_3 10}}$ $\therefore\ b > a$

(iv) $a = \log_{\frac{1}{5}} \frac{1}{7}$ $b = \log_{\frac{1}{7}} \frac{1}{5}$

$a = \log_5 7$ $b = \log_7 5$

$a > 1$ $b < 1$

$\therefore\ a > b$

22. If the product of all solutions of the equation $\frac{(2009)x}{2010} = (2009)^{\log_x(2010)}$ can be expressed in the lowest form as $\frac{m}{n}$ then the value of $(m+n)$ is.

यदि समीकरण $\frac{(2009)x}{2010} = (2009)^{\log_x(2010)}$ के सभी मूलो का गुणनफल यदि $\frac{m}{n}$ के रूप में व्यक्त किया, जहाँ m व n का महत्तम समापर्वतक 1 है, तो m + n का मान होगा।

Ans. 4019 where $\frac{m}{n} = \frac{2010}{2009}$

Sol. $\frac{(2009)x}{2010} = (2009)\log_x(2010)$

taking log of both sides with base 2009

$$\log_{2009}\frac{2009x}{2010} = \log_x 2010 \cdot \log_{2009} 2009$$

$$\log_{2009}2009x - \log_{2009}2010 = \log_x 2010$$

$$\log_{2009}2009 + \log_{2009}x - \log_{2009}2010 - \log_x 2010$$

$$1 + \log_{2009} x - \log_{2009} 2010 = \frac{\log_{2009} 2010}{\log_{2009} x}$$

put $\log_{2009}x = t$

$$1 + t - \log_{2009} 2010 = \frac{\log_{2009} 2010}{t}$$

$$t^2 + (1 - \log_{2009}2010)t - \log_{2009}2010 = 0$$

$$t = \frac{-(1-\log_{2009} 2010) \pm \sqrt{(1-\log_{2009} 2010)^2 + 4\log_{2009} 2010}}{2}$$

$$t = \frac{-(1-\log_{2009} 2010) \pm (1+\log_{2009} 2010)}{2}$$

for +ve

$$t = \frac{-(1-\log_{2009} 2010) + (1+\log_{2009} 2010)}{2}$$

$$t = \frac{1}{2}[-1 + \log_{2009} 2010 + 1\log_{2009} 2010] = \log_{2009} 2010$$

$$\Rightarrow \log_{2009}x = \log_{2009}2010 \Rightarrow x = 2010$$

for –ve

$$t = \frac{-1 + \log_{2009} 2010 - 1 - \log_{2009} 2010}{2}$$

$t = -1$

$\log_{2009} x = -1$

$x = 2009^{-1}$

$$x = \frac{1}{2009} = x_1$$

$$\text{Product of roots} = \frac{2010}{2009} = x_1 \times x_2 = \frac{m}{n}$$

$\therefore\ m + n = 2010 + 2009$

$= 4019$

23. If p, q are the roots of the quadratic equation $x^2 + 2bx + c = 0$, prove that

$$2\log\left(\sqrt{y-p} + \sqrt{y-q}\right) = \log 2 + \log\left(y + b + \sqrt{y^2 + 2by + c}\right)$$

यदि p, q द्विघात समीकरण $x^2 + 2bx + c = 0$ के मूल है तो सिद्ध कीजिये

$$2\log\left(\sqrt{y-p} + \sqrt{y-q}\right) = \log 2 + \log\left(y + b + \sqrt{y^2 + 2by + c}\right)$$

Ans.

Sol. $x^2 + 2bx + c = 0 \begin{matrix} p \\ q \end{matrix}$; $\left.\begin{matrix} p+q = -2b \\ pq = c \end{matrix}\right\}$ ....(1)

TPT $2\log\left(\sqrt{y-p} + \sqrt{y-q}\right) = \log 2 + \log\left(y + b + \sqrt{y^2 + 2by + c}\right)$

$$\text{LHS} = \log\left(y - p + y - q + 2\sqrt{y-p}\sqrt{y-q}\right)$$

$$= \log\left(2y - (p+q) + 2\sqrt{y^2 - (p+q)y + pq}\right)$$

$$= \log\left(2y + 2b + 2\sqrt{y^2 + 2by + c}\right) = \log 2 + \log\left(y + b + \sqrt{y^2 + 2by + c}\right) = \text{RHS}$$

24. The number of solutions of the equation $\log_4(x-1) = \log_2(x-3)$ is : [JEE Main Feb 2021]

समीकरण $\log_4(x-1) = \log_2(x-3)$ के हलों की संख्या है _____

Ans. (1)

Sol. $\log_4(x-1) = \log_2(x-3)$

$x - 1 > 0$ and $x - 3 > 0 \Rightarrow x > 3$

$$\log_2(x-1) = \log_2(x-3) \Rightarrow \frac{1}{2}\log_2(x-1) = \log_2(x-3)$$

$\Rightarrow \log_2(x-1) = 2\log_2(x-3) = \log_2(x-3)^2$

$\Rightarrow x-1 = (x-3)^2$

$\Rightarrow x^2 - 7x + 10 = 0$

$x = 2, 5$

$\Rightarrow x = 2$(Rejected ; as $x > 3$)

So no. of values = 1

25. Find the sum of the roots of the equation $2^{333x-2} + 2^{111x+1} = 2^{222x+2} + 1$.

समीकरण $2^{333x-2} + 2^{111x+1} = 2^{222x+2} + 1$ के मूलो का योग क्या होगा।

Ans. $\frac{2}{111}$

Sol. Let $2^{111x} = y \Rightarrow \log_2 y = 111x$

so that $\log_2 y = 111\,x \Rightarrow x = \frac{\log_2 y}{111}$

equation becomes

$\frac{y^3}{4} + 2y = 4y^2 + 1$

$y^3 - 16y^2 + 8y - 4 = 0$

sum of the roots of the given equation is

$x_1 + x_2 + x_3 = \frac{\log_2 y_1 + \log_2 y_2 + \log_2 y_3}{111} = \frac{\log_2(y_1y_2y_3)}{111} = \frac{\log_2 4}{111} = \frac{2}{111}$ Ans.

26. The value of $\left((\log_2 9)^2\right)^{\frac{1}{\log_2(\log_2 9)}} \times \left(\sqrt{7}\right)^{\frac{1}{\log_4 7}}$ is _____. [JEE(Adv)-2018]

$\left((\log_2 9)^2\right)^{\frac{1}{\log_2(\log_2 9)}} \times \left(\sqrt{7}\right)^{\frac{1}{\log_4 7}}$ का मान है _____. [JEE(Adv)-2018]

Ans. 8

Sol. $\left((\log_2 9)^2\right)^{\frac{1}{\log_2(\log_2 9)}} \times \left(\sqrt{7}\right)^{\frac{1}{\log_4 7}}$

$= \left((\log_2 9)^2\right)^{\log_{\log_2 9} 2} \times 7^{\frac{1}{2}\log_7 4}$

$= \left(2^2\right)^{\log_{\log_2 9}\log_2 9} \times \left(4^{\frac{1}{2}}\right)^{\log_7 7}$

$= 4 \times 2 = 8$

27. For $x \in R$, the number of real roots of the equation $3x^2 - 4|x^2-1| + x - 1 = 0$ is ___. [JEE(Adv)-2021]

$x \in R$ के लिए, समीकरण $3x^2 - 4|x^2-1| + x - 1 = 0$ के वास्तविक मूलों की संख्या ___ है। [JEE(Adv)-2021]

Ans. 4

Sol. $3x^2 - 4|x^2 - 1| + x - 1 = 0$

Case I: $x^2 - 1 \geq 0$

$\Rightarrow x \in (-\infty, -1] \cup [1, \infty)$ .....(1)

$3x^2 - 4(x^2 - 1) + x - 1 = 0$

$3x^2 - 4x^2 + 4 + x - 1 = 0$

$x^2 - x - 3 = 0$

$\Rightarrow x = \frac{1 \pm \sqrt{13}}{2}$ (Both are acceptable)

Case II: $x^2 - 1 \leq 0$

$x \in [-1, 1]$ .....(2)

$3x^2 + 4(x^2 - 1) + x - 1 = 0$

$7x^2 + x - 5 = 0$

$\Rightarrow x = \frac{-1 \pm \sqrt{141}}{14}$ (Both belongs to (2))

Total solution = 4

28. Match the following columns :

	Column-I	**Column-II**
A.	The value of K, where $\log(\log 4) + \log(\log 25) = \log K + \log(\log 2) + \log(\log 5)$ is	p. 1
B.	Number of values of $x \in N$, for which $x^4 + 4$ is prime, is	q. 2
C.	If b is a positive real number different from 1, let $\log_b x$ denotes the base b logarithm of x. Let n be the number of solutions of x to the equation, $\log_b x = \log_x b$ where x is a positive real different from 1. Then n equals	r. 3
D.	The expression $\sqrt{\log_{0.5}^2 8}$ has the value equal to _____.	s. 4

निम्न Columns को सुमेलित कीजिए :

	Column-I	**Column-II**
A.	यदि $\log(\log 4) + \log(\log 25) = \log K + \log(\log 2) + \log(\log 5)$	p. 1

तो K का मान होगा

B. x के कितने मानो के लिये $x^4 + 4$ एक अभाज्य संख्या होगी जहां $x \in N$ — q. 2

C. यदि b एक धनात्मक इकाई ये भिन्न वास्तविक संख्या है, माना समीकरण $\log_b x = \log_x b$ के n हल है जहाँ पर x एक धनात्मक इकाई ये भिन्न वास्तविक संख्या है तो n का मान होगा — r. 3

D. व्यजंक $\sqrt{\log_{0.5}^2 8}$ का मान बराबर होगा — s. 4

Ans. (A) S ; (B) P ; (C) Q; (D) R

Sol. (A) $\log(\log 4) + \log(\log 25) = \log K + \log(\log 2) + \log(\log 5)$

$\log(2\log 2) + \log(2\log 5) = \log k + \log(\log 2) + \log(\log 5)$

$\log 2 + \log 2 = \log K$

$\log 4 = \log K \Rightarrow K = 4$ Ans.

(B) $x^4 + 4 = (x^4 + 4x^2 + 4) - 4x^2 = (x^2 + 2)^2 - (2x)^2$; $x = 1$ Ans.

(C) $\log_b x = \dfrac{1}{\log_b x}$

$(\log_b x)^2 = 1 \Rightarrow \log_b x = 1$ or -1

$x = b$ or $\dfrac{1}{b} \Rightarrow n = 2$ Ans.

(D) Note $\sqrt{x^2} = -x$, if x is negative

$$\sqrt{\log_{0.5}^2 8} = \sqrt{(-3\log_2 2)^2} = \sqrt{(-3)^2} = 3$$

29. Match the following columns :

Column-I	**Column-II**
A. If x_1 and x_2 satisfy the equation $x^{\log_{10} x} = 100x$ then the value of x_1x_2 equals	p. irrational
B. Sum of the squares of the roots of the equation $\log_2(9 - 2^x) = 3 - x$ is	q. rational
C. If $\log_{1/8}\left(\log_{1/4}\left(\log_{1/2} x\right)\right) = \dfrac{1}{3}$ then x is	r. prime
D. If $\log_b a = 3$, $\log_b c = -4$. If the value of x satisfying the equation $a^{3x} = c^{x-1}$ is expressed in the form p/q where p and q are relatively prime then p + q is	s. composite

निम्न Columns को सुमेलित कीजिए :

Column-I	**Column-II**

A.	यदि x_1 व x_2 समीकरण $x^{\log_{10} x} = 100x$ को दो मूल हो तो x_1x_2 का मान होगा	p.	अपरिमेय
B.	समीकरण $\log_2(9 - 2^x) = 3 - x$ के मूलो के वर्गो का योग होगा	q.	परिमेय
C.	यदि $\log_{1/8}\left(\log_{1/4}\left(\log_{1/2} x\right)\right) = \frac{1}{3}$ तो x है	r.	अभाज्य
D.	यदि $\log_b a = 3$, $\log_b c = -4$ समीकरण $a^{3x} = c^{x-1}$ को सन्तुष्ट करने वाले x का मान यदि p/q के रूप में व्यक्त हो, जहां पर p व q का महत्तम समापवर्तक 1 हो, तो p + q का मान होगा	s.	मिश्र

Ans. (A) Q, S; (B) Q, S; (C) P; (D) Q, R

Sol. (A) $\log_{10}^2 x = \log_{10} 100x = 2 + \log_{10} x$

put $\log x = t$

$t^2 = 2 + t \quad \Rightarrow \quad t^2 - t - 2 = 0 \quad \Rightarrow \quad (t-2)(t+1) = 0$

$\Rightarrow \quad t = 2 \quad \text{or} \quad t = -1$

$\log_{10} x = 2 \quad \text{or} \quad \log_{10} x = -1$

$x = 100 \quad \text{or} \quad x = 1/10$

(B) $\log_2(9 - 2^x) = 3 - x$

$2^{(3-x)} = 9 - 2^x$

$$\frac{8}{2^x} = 9 - 2^x$$

put $2^x = t$,

$8 = 9t - t^2 \quad \Rightarrow \quad t^2 - 9t + 8 = 0 \quad \Rightarrow \quad (t-8)(t-1) = 0$

$t = 8 \quad \text{or} \quad t = 1$

$2^x = 8 \quad \text{or} \quad 2^x = 1$

$x = 3 \quad \text{or} \quad x = 0$

(C) If $\log_{1/8}\left(\log_{1/4}\left(\log_{1/2} x\right)\right) = \frac{1}{3}$

$$\left(\frac{1}{8}\right)^{1/3} = \log_{1/4}\left(\log_{1/2} x\right); \qquad \frac{1}{2} = \log_{1/4}\left(\log_{1/2} x\right); \quad \left(\frac{1}{4}\right)^{1/2} = \log_{1/2} x; \quad \frac{1}{2} = \log_{1/2} x$$

$$\left(\frac{1}{2}\right)^{1/2} = x; \qquad \therefore \qquad x = \frac{1}{\sqrt{2}}$$

(D) Given $\log_b a = 3$, $\log_b c = -4$

Given, $a^{3x} = c^{x-1}$

$3x \log_b a = (x-1) \log_b c$

$3x \times 3 = -4(x-1) \quad \Rightarrow \quad 9x = 4 - 4x \quad \Rightarrow \quad x = 4/13$